Variant calling & Annotation

Massimo Maiolo

2.10.2025

Institute of Tissue Medicine and Pathology – University of Bern
Clinical Genomics Lab - **INSEL**SPITAL

Github Cancer Genomics Course

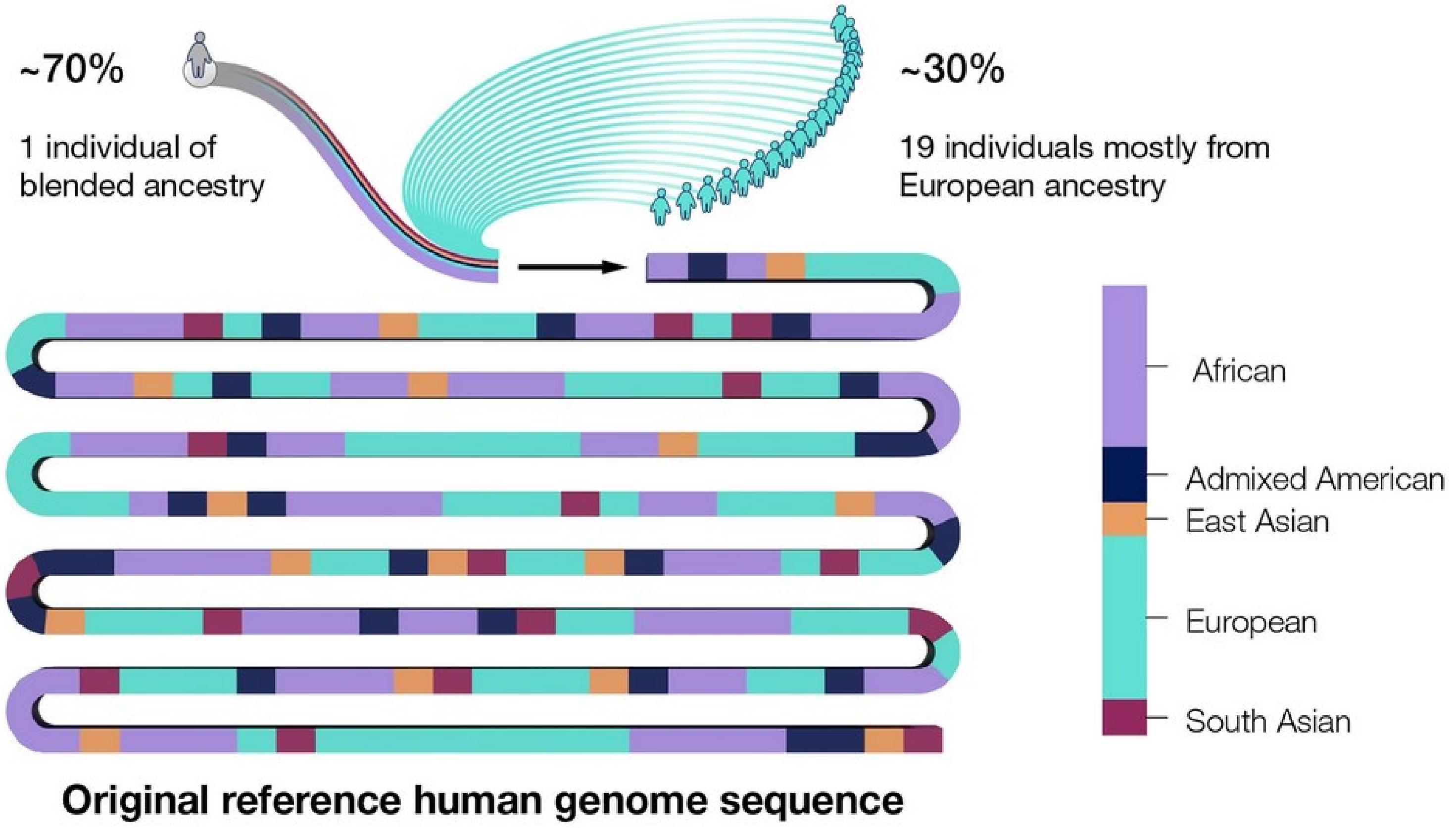
https://github.com/supermax1234/CancerGenomicsCourse

Comparing individual human genome sequences

https://storymd.com

The Reference Genome: A Standardized Map

The reference genome serves as a **universal map** for comparing genomic sequences. It is a **composite, haploid sequence** representing the **most common alleles** across a population. It's crucial to understand that it's **not a single individual's genome** but a standardized template that provides a common coordinate system.



Detecting variants in a person's genome sequence usually requires an existing sequence for comparison — a *reference*.

A reference human genome sequence is an *established*, *high-quality* and *well-accepted sequence* of a human genome (i.e., one sequence for each of the 23 human chromosomes).

A reference human genome sequence is not an actual sequence from one individual but pieced together from multiple people. The important feature of a reference human genome sequence is that it depicts one assigned nucleotide for every position across the human genome. In this regard, a reference human genome sequence only represents the sequence of one copy of each chromosome, whereas a person's genome sequence contains the sequences of both copies of each chromosome (i.e., people are diploid).

The Reference Genome: A Standardized Map

- **The Human Genome Reference Sequence:**
  - **Current Standard (de-facto): hg19 / GRCh37** : These two versions are functionally the same and are still very widely used, especially in older datasets and clinical pipelines. Many established tools and databases (like dbSNP and 1000 Genomes) are built on this assembly.
  - **Newer Standard: GRCh38**: This is the current, more complete reference assembly. It includes improvements such as more accurate centromeric regions and the incorporation of *alternative loci* to better represent human genetic diversity. All major consortia (like the Genome Aggregation Database, gnomAD) now use GRCh38. The transition from hg19 to hg38 requires a process called *liftover*, which can be a significant technical challenge.
- **Other Organisms:** Not all organisms have a fully assembled reference genome.

Liftover

Lift over variants, positions, or intervals from one reference genome to another.

This web interface supports more input formats and requires fewer clicks than existing interfaces like UCSC LiftOver and Ensembl Assembly Converter.

Under the hood, it uses the UCSC liftOver command-line tool for **intervals** and **positions**, while for **variants** it uses a new bcftools plugin which avoids many of the limitations of other liftover tools (particularly for indels) as described in [Genovese 2024].

To submit issues or feature requests, please use broadinstitute/liftover/issues.

Input formats:

- **Positions:**
  chr8:141310715
  chr8-141310715
  chr8 141310715
- **Variants:**
  chr8:141310715 T>G
  chr8-141310715-T-G
  NM_001089.3(ABCA3):c.875A>T (p.Glu292Val)
- **Intervals:**
  chr8:141310715-141310720
  chr8 141310715 141310720

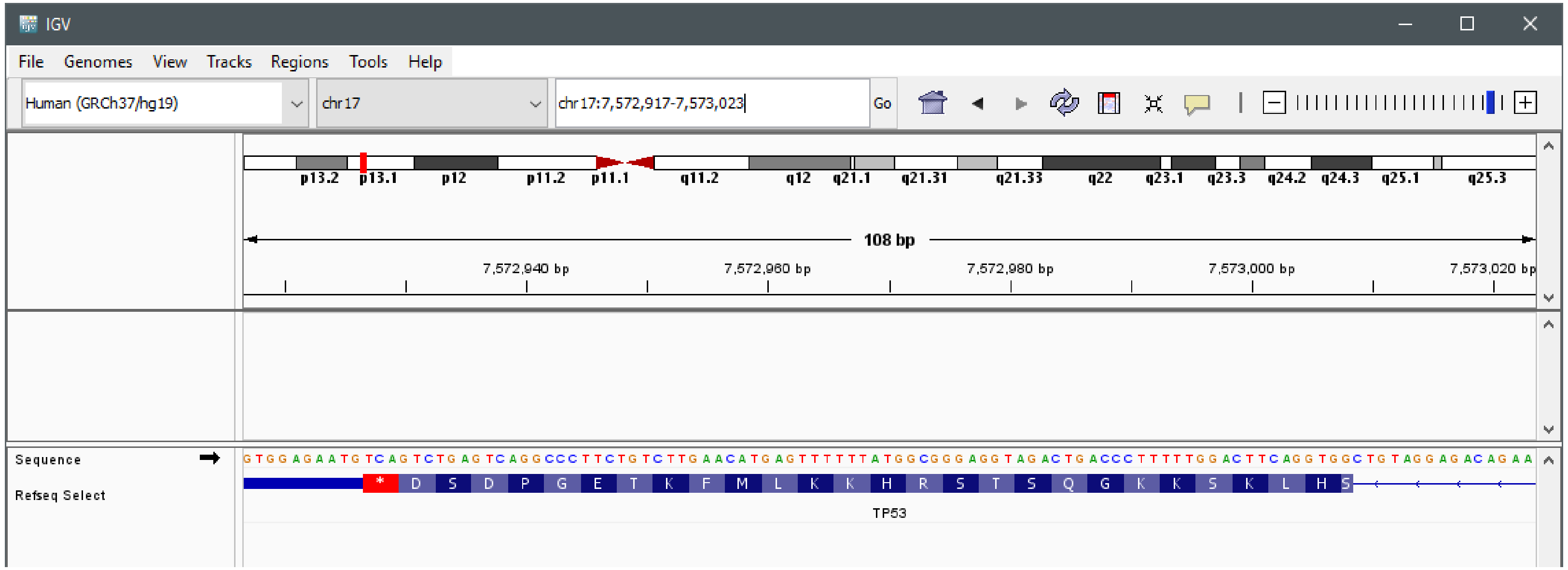
The "chr" prefix is optional, and separators can be `:` , `-` , `>` , or space(s).

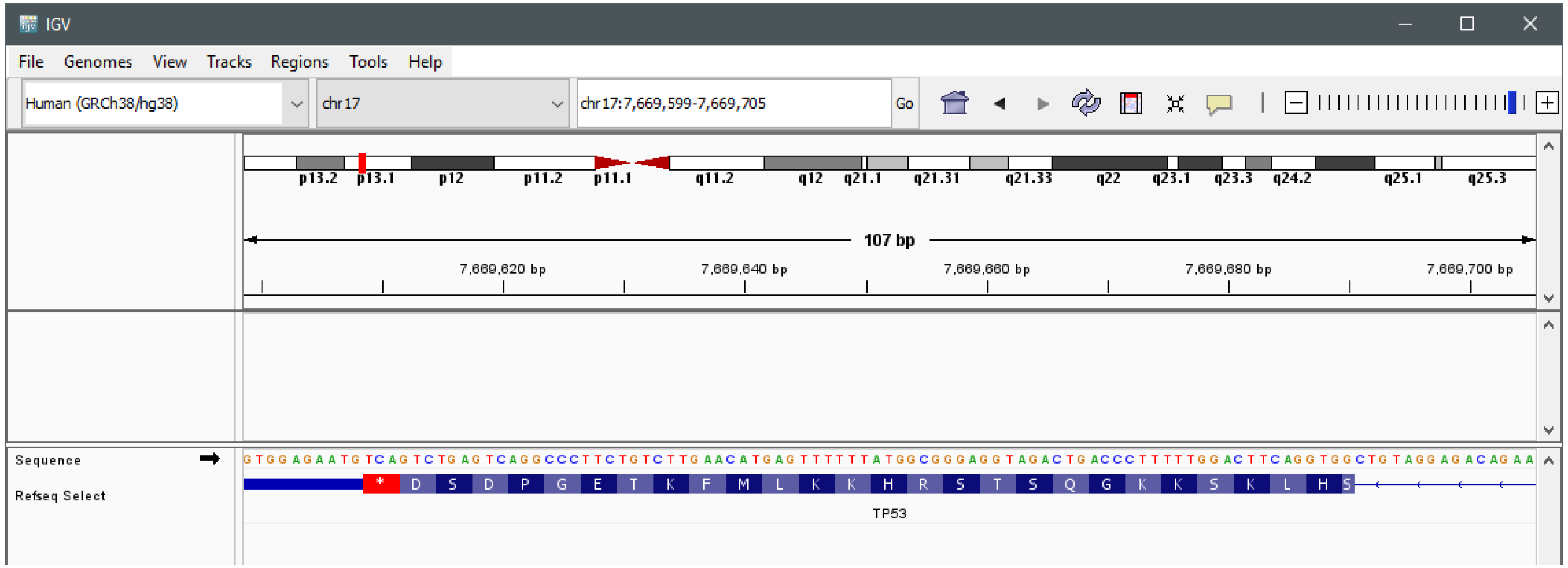
Enter a position, variant, or interval

( ) hg19 => hg38 ( ) hg38 => hg19 Submit

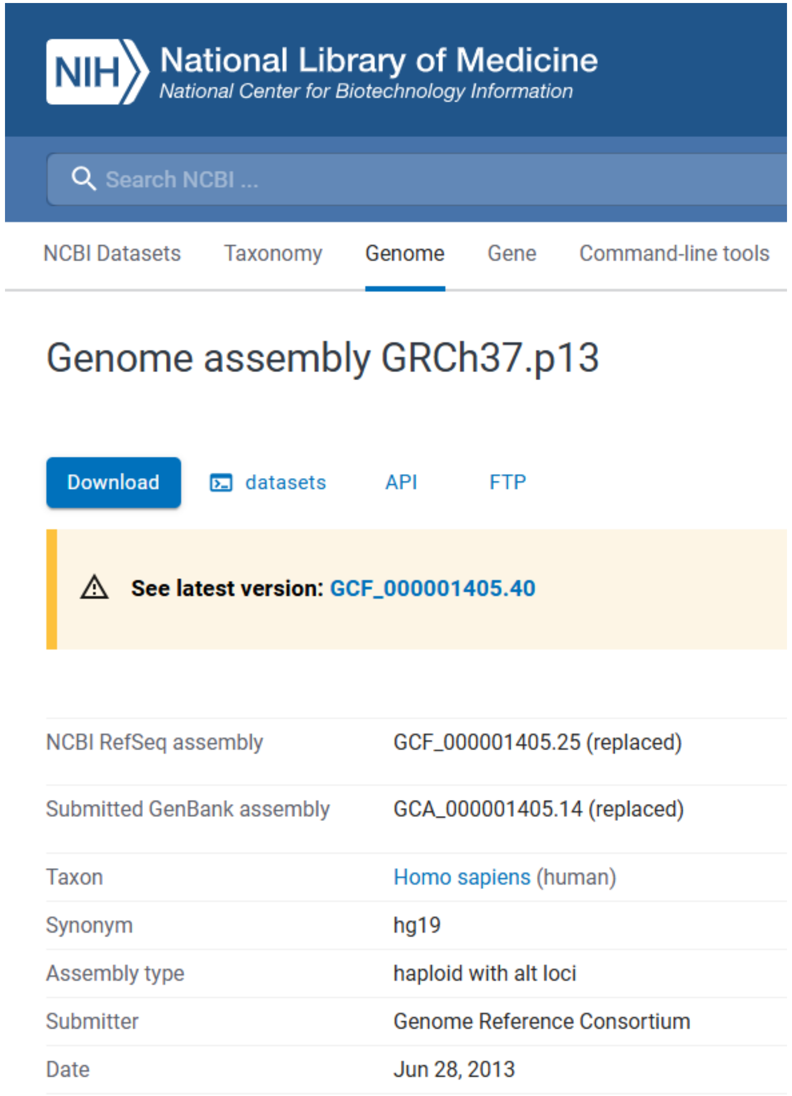
( ) hg38 => T2T ( ) T2T => hg38

https://liftover.broadinstitute.org/#



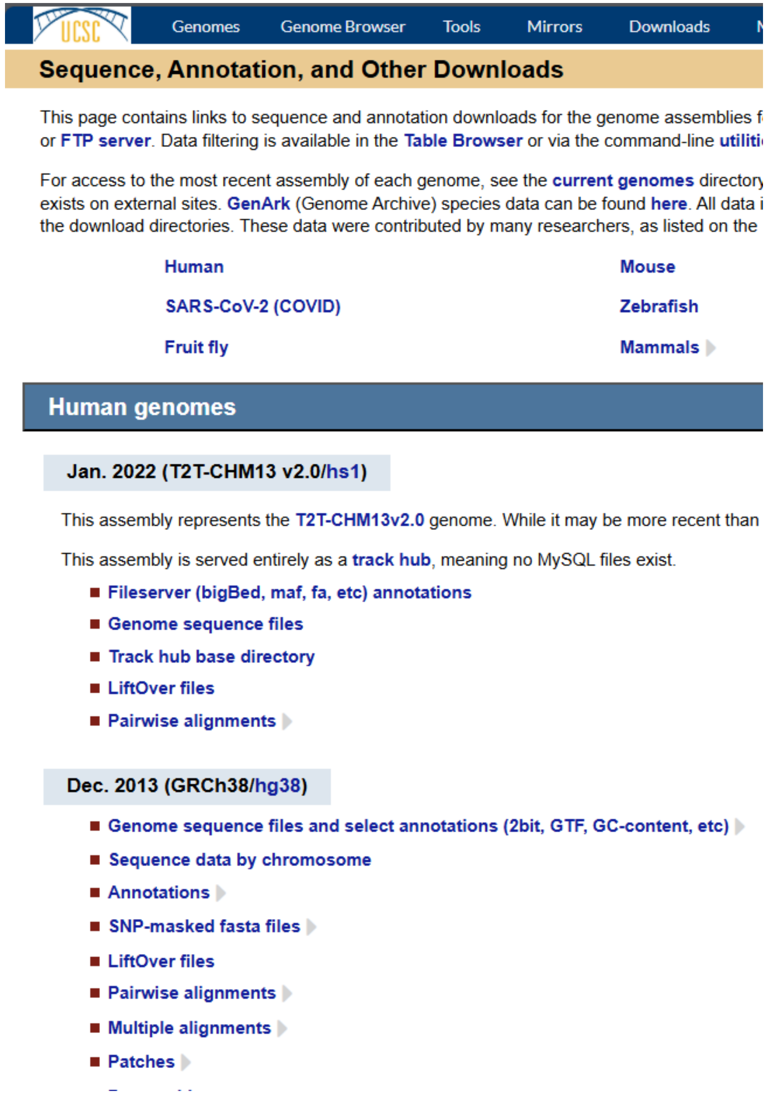


National Center for Biotechnology Information (NCBI)



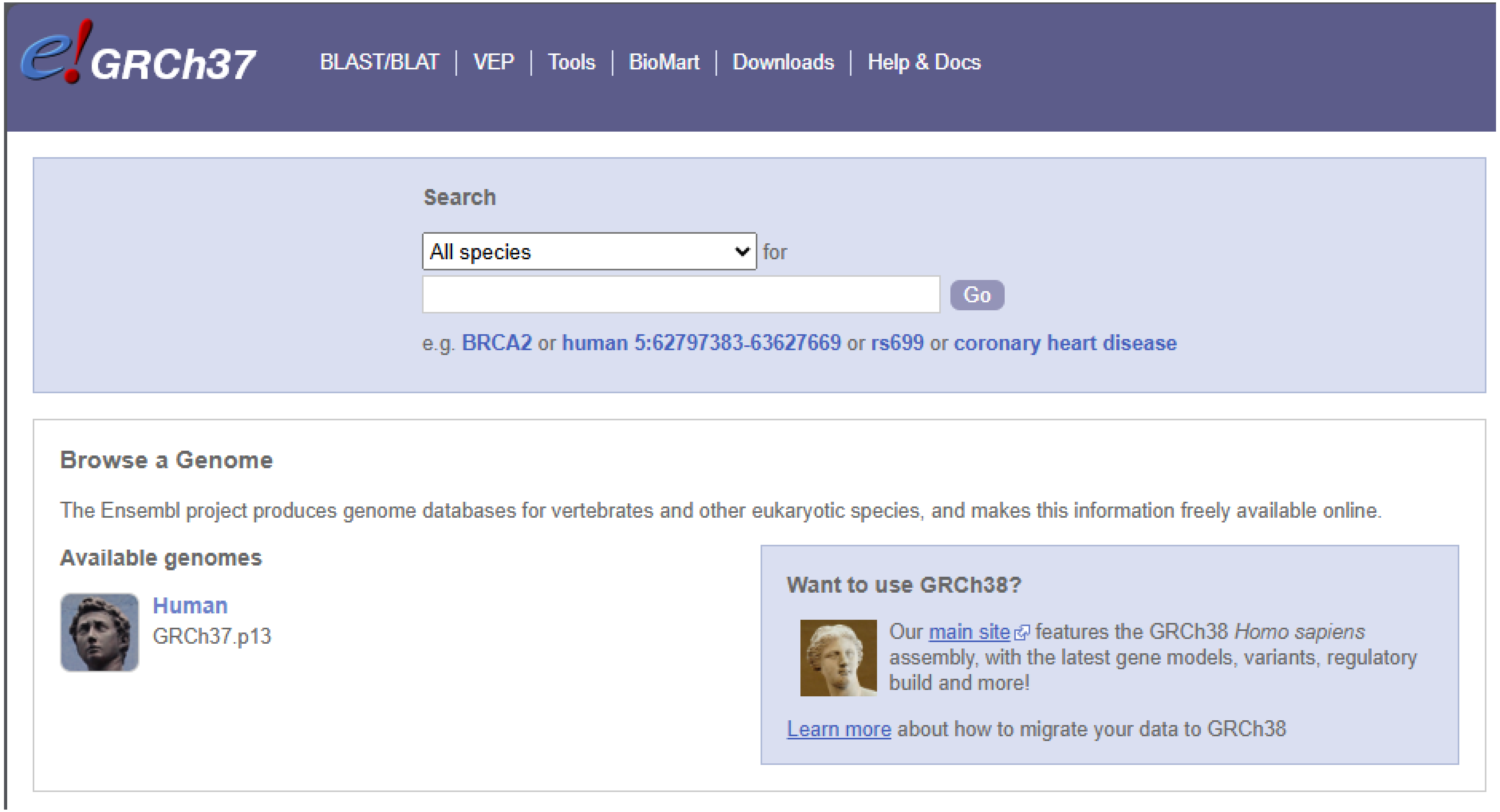
https://www.ncbi.nlm.nih.gov/assembly/37871

UCSC Genome Browser



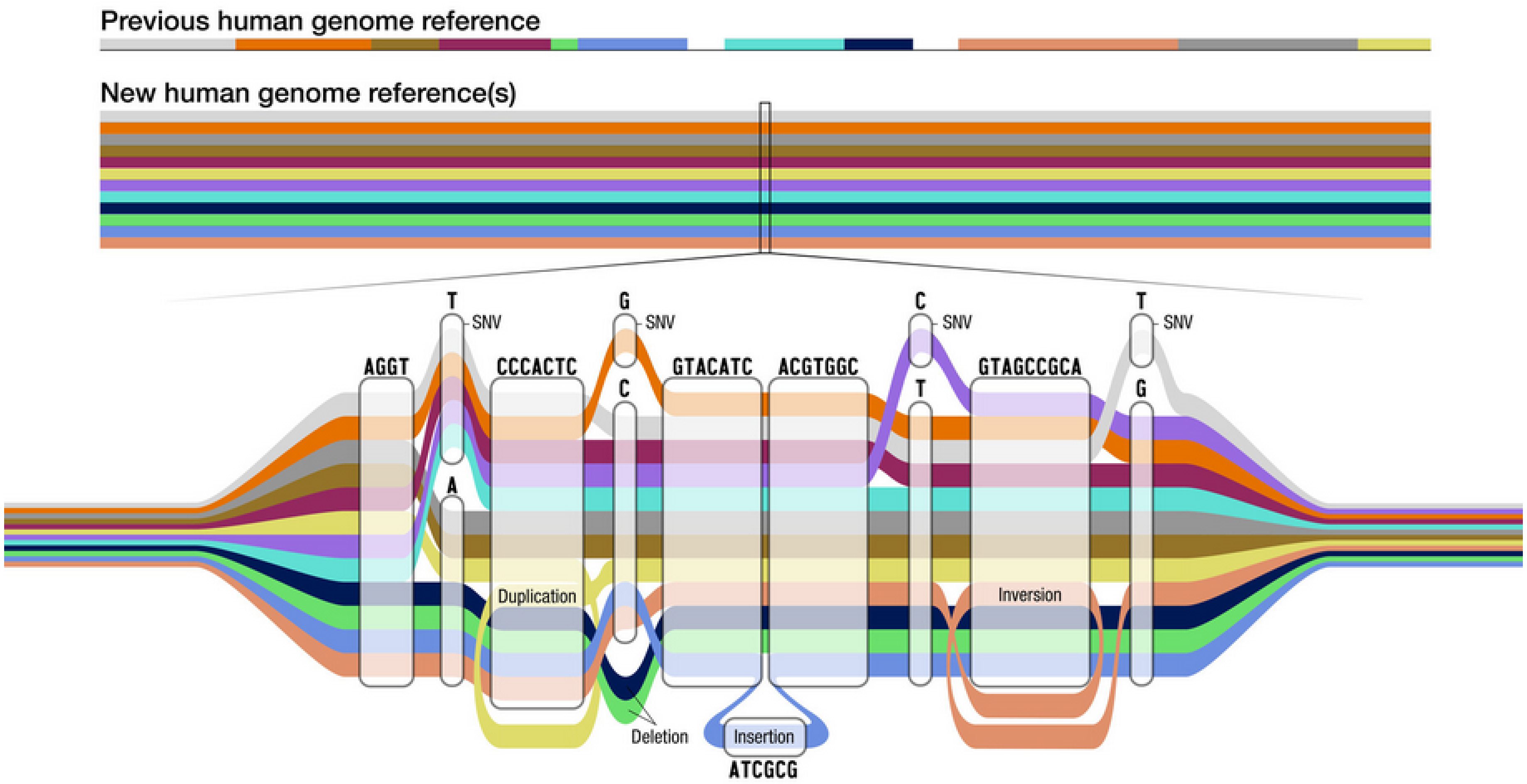
https://hgdownload.soe.ucsc.edu/downloads.html#human

Alternative Site: Ensembl



https://grch37.ensembl.org/

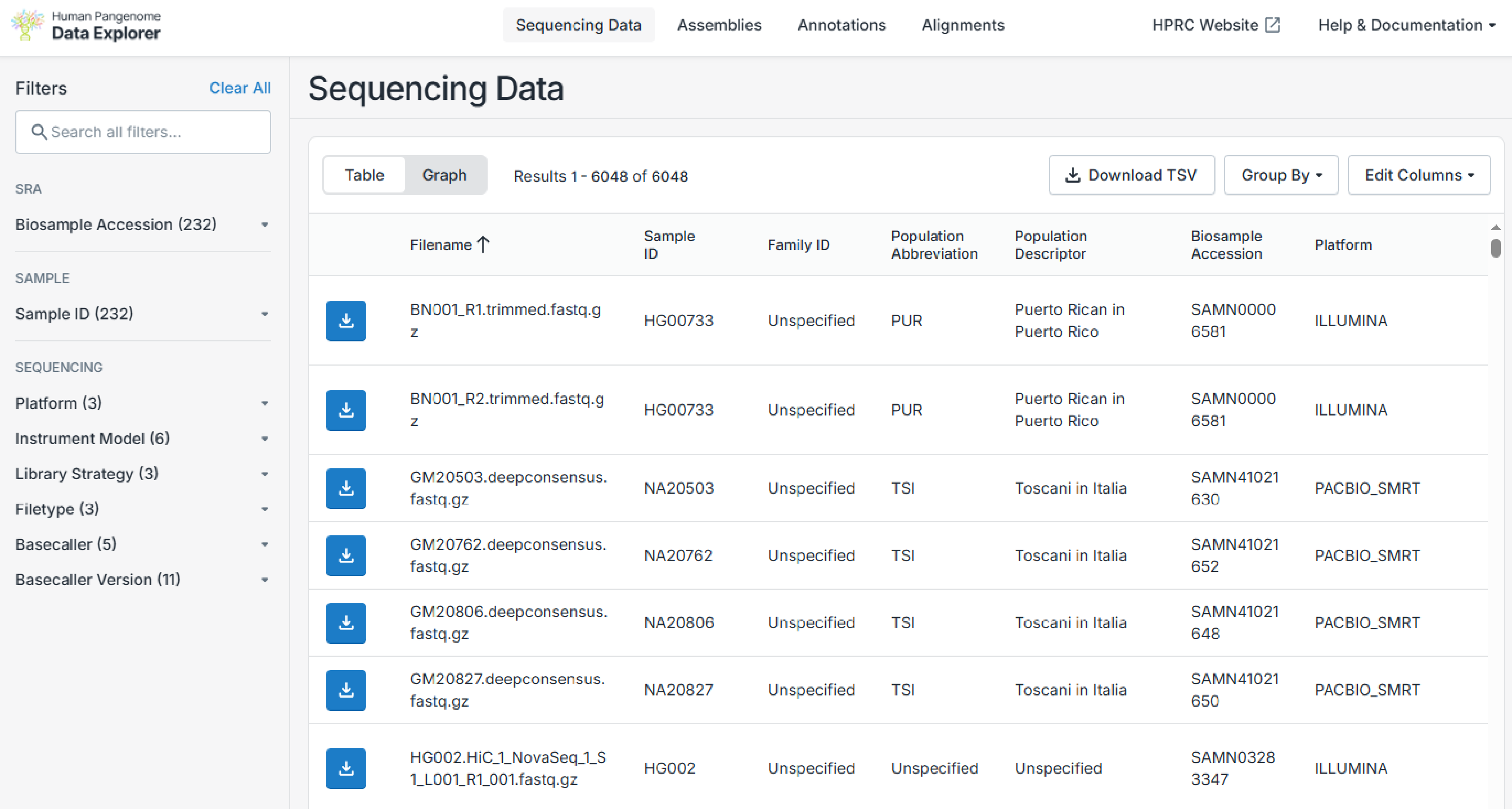
What Is a Pangenome Reference Sequence and How Is It Used?



Reference human genome sequences are invaluable for detecting genomic variants in each newly sequenced human genome. But the currently available reference human genome sequences do not accurately represent the genomic diversity of the human population, and that lack of **diversity** can introduce **biases** when analyzing some peoples' genome sequences.

To address this deficiency, researchers are working to generate a more complete set of reference human genome sequences that better reflect all of humanity. A "pangenome" is the collective genome sequences of multiple individuals that better represents the genomic diversity of the species. The human pangenome reference sequence will provide a better tool for comparing genome sequences from people all over the world in an effort to detect and characterize genomic variants more completely, including those with important roles in health and disease.

Human Pangenome Reference Consortium (HPRC)


Human Pangenome Data Explorer

Sequencing Data | Assemblies | Annotations | Alignments | HPRC Website | Help & Documentation

Filters — Clear All

Search all filters...

SRA
- Biosample Accession (232)

SAMPLE
- Sample ID (232)

SEQUENCING
- Platform (3)
- Instrument Model (6)
- Library Strategy (3)
- Filetype (3)
- Basecaller (5)
- Basecaller Version (11)

Sequencing Data

Table | Graph — Results 1 - 6048 of 6048 — Download TSV | Group By | Edit Columns

Filename	Sample ID	Family ID	Population Abbreviation	Population Descriptor	Biosample Accession	Platform
BN001_R1.trimmed.fastq.gz	HG00733	Unspecified	PUR	Puerto Rican in Puerto Rico	SAMN00006581	ILLUMINA
BN001_R2.trimmed.fastq.gz	HG00733	Unspecified	PUR	Puerto Rican in Puerto Rico	SAMN00006581	ILLUMINA
GM20503.deepconsensus.fastq.gz	NA20503	Unspecified	TSI	Toscani in Italia	SAMN41021630	PACBIO_SMRT
GM20762.deepconsensus.fastq.gz	NA20762	Unspecified	TSI	Toscani in Italia	SAMN41021652	PACBIO_SMRT
GM20806.deepconsensus.fastq.gz	NA20806	Unspecified	TSI	Toscani in Italia	SAMN41021648	PACBIO_SMRT
GM20827.deepconsensus.fastq.gz	NA20827	Unspecified	TSI	Toscani in Italia	SAMN41021650	PACBIO_SMRT
HG002.HiC_1_NovaSeq_1_S1_L001_R1_001.fastq.gz	HG002	Unspecified	Unspecified	Unspecified	SAMN03283347	ILLUMINA

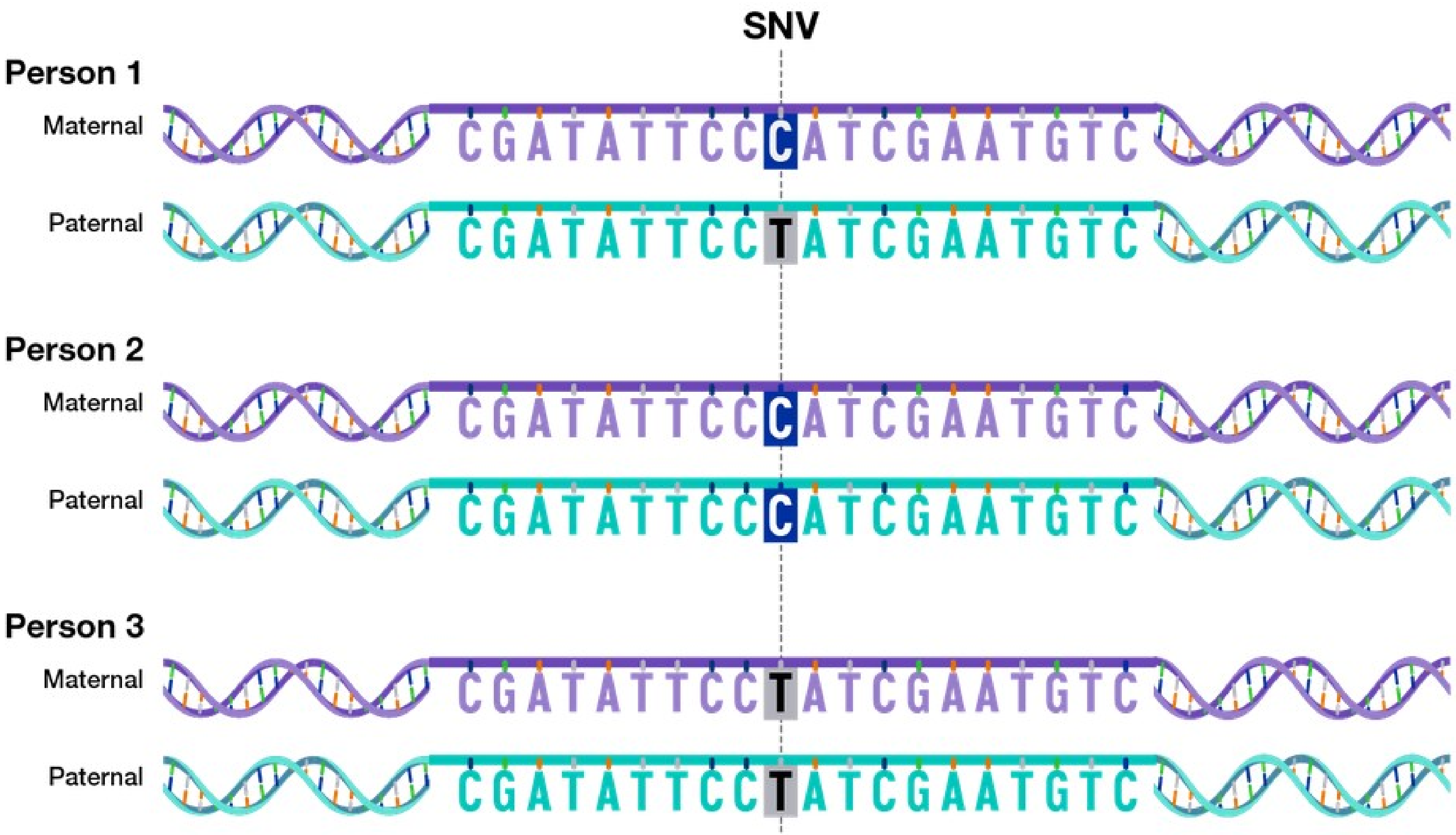
https://data.humanpangenome.org/

Why Variation Matters?

The vast **majority of the DNA letters** in peoples' genomes is **identical**, but a small fraction of those letters varies. This **genomic variation** accounts for some of the differences among people, including important aspects of their *health* and *susceptibility to diseases*.

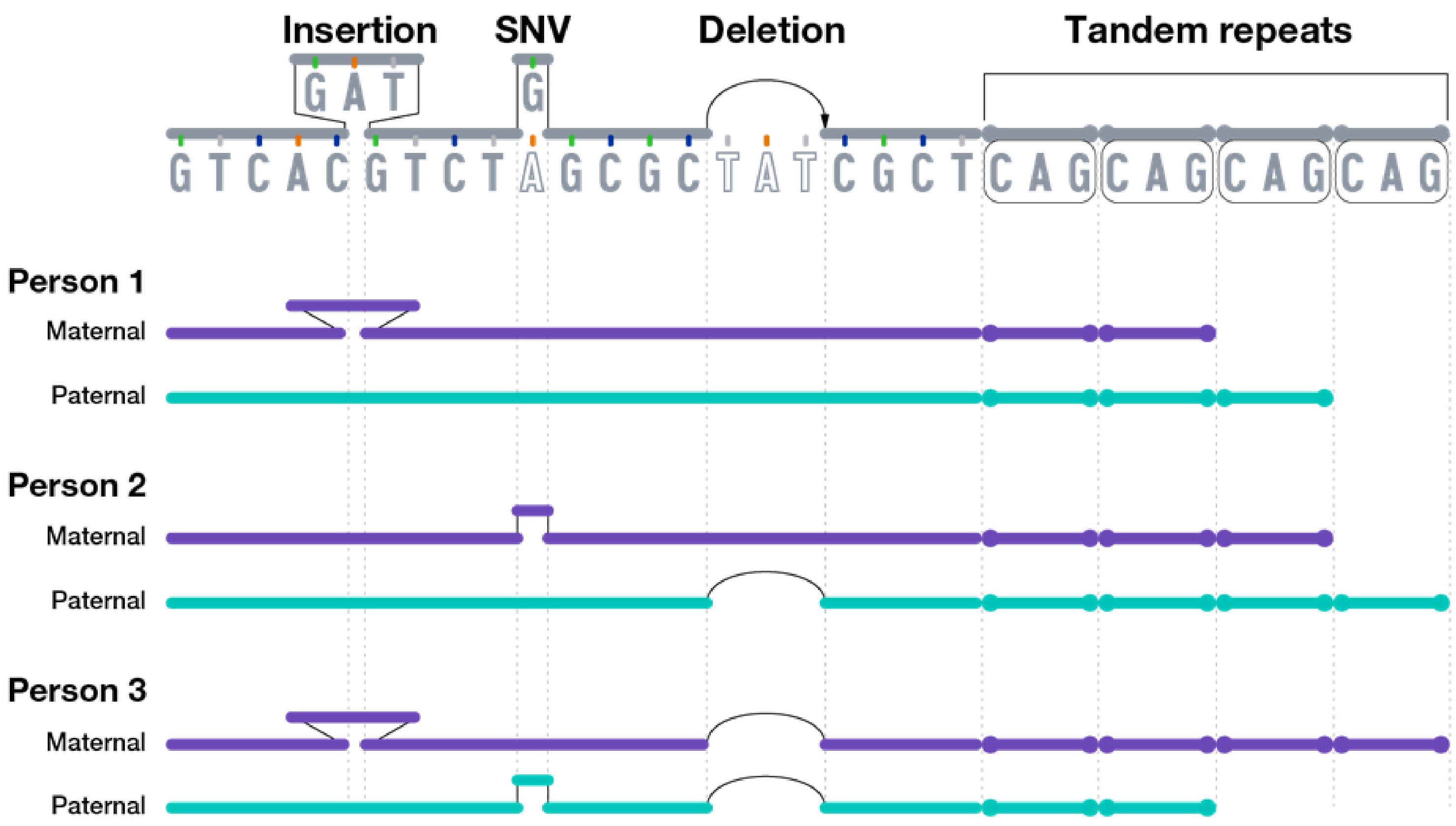
- **Genomic variation** reflects the differences in a person's DNA compared to other peoples' DNA.
- There are multiple types of variants in human genomes, ranging from ***small*** differences to ***large*** differences.
- A very small subset of genomic variants contributes to human health and disease.
- Researchers create reference **human genome** sequences to help **detect genomic variants** in each sequenced human genome.
- On average, a person's genome sequence is **~99.6% identical** to a reference **human genome** sequence; that person's set of genomic variants accounts for the **~0.4% difference**.
- The *human pangenome* is a more comprehensive framework that aims to account for **genomic variation** across human populations, thereby reducing biases that can come with the use of a single reference **human genome** sequence.

What Are the Different Types of Genomic Variants?



The smallest genomic variants are **single-nucleotide variants** (SNVs). Each SNV reflects a difference in a single nucleotide. For a given SNV, the DNA letter at that genomic position might be a C in one person but a T in another person. SNVs are the most common type of **genomic variation**. A subtype of SNVs is called a single-nucleotide poly-morphism (SNP; pronounced "snip"). To be considered a SNP, a SNV must be present in **at least 1% of the human population**. As such, SNV is a more general term that includes both relatively common (such as SNPs) and rare single-nucleotide differences. For simplicity, we refer to all single-nucleotide differences as SNVs, regardless of their relative frequency.

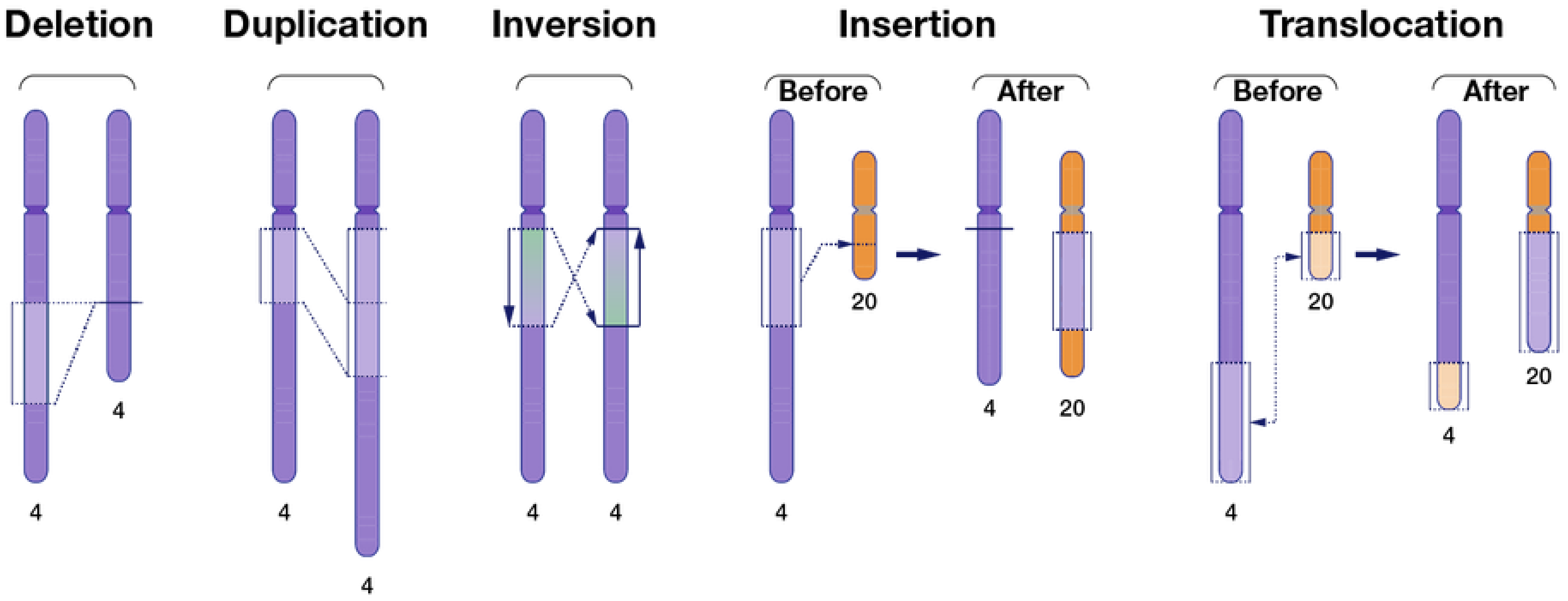
What Are the Different Types of Genomic Variants?



Another group of small genomic variants are **insertions** and **deletions** (*indels*). Insertion/deletion variants reflect extra or missing DNA nucleotides in the genome, respectively, and typically involve fewer than 50 nucleotides. Insertion/deletion variants are less frequent than SNVs but can sometimes have a larger impact on health and disease.

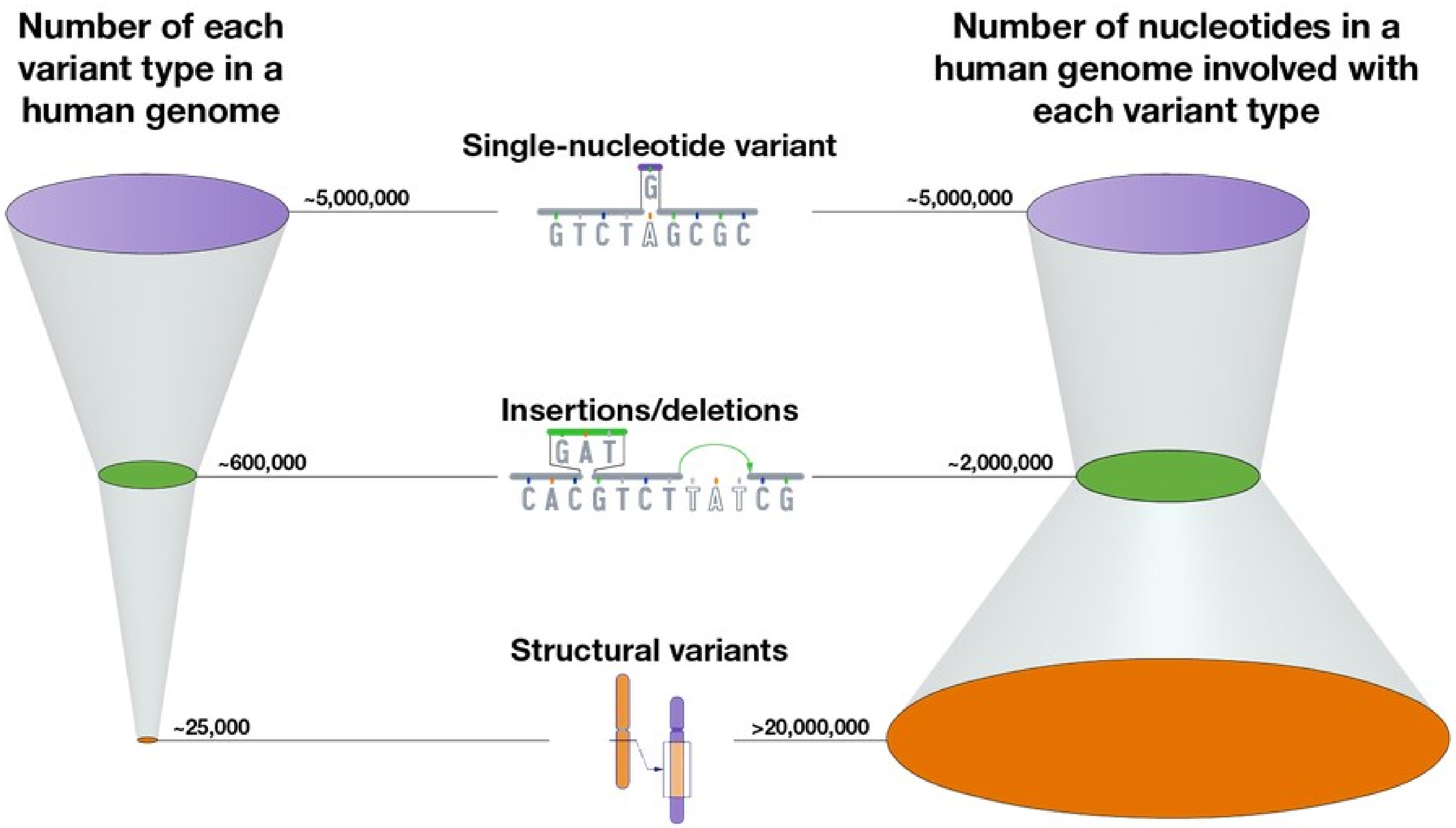
One of the most common types of insertion/deletion variants are tandem repeats (also known as **microsatellites**). Tandem repeats are short stretches of nucleotides that are repeated multiple times and are highly variable among people. Different **chromosomes** can vary in the number of times such short nucleotide stretches are repeated, ranging from a few times to hundreds of times.

What Are the Different Types of Genomic Variants?



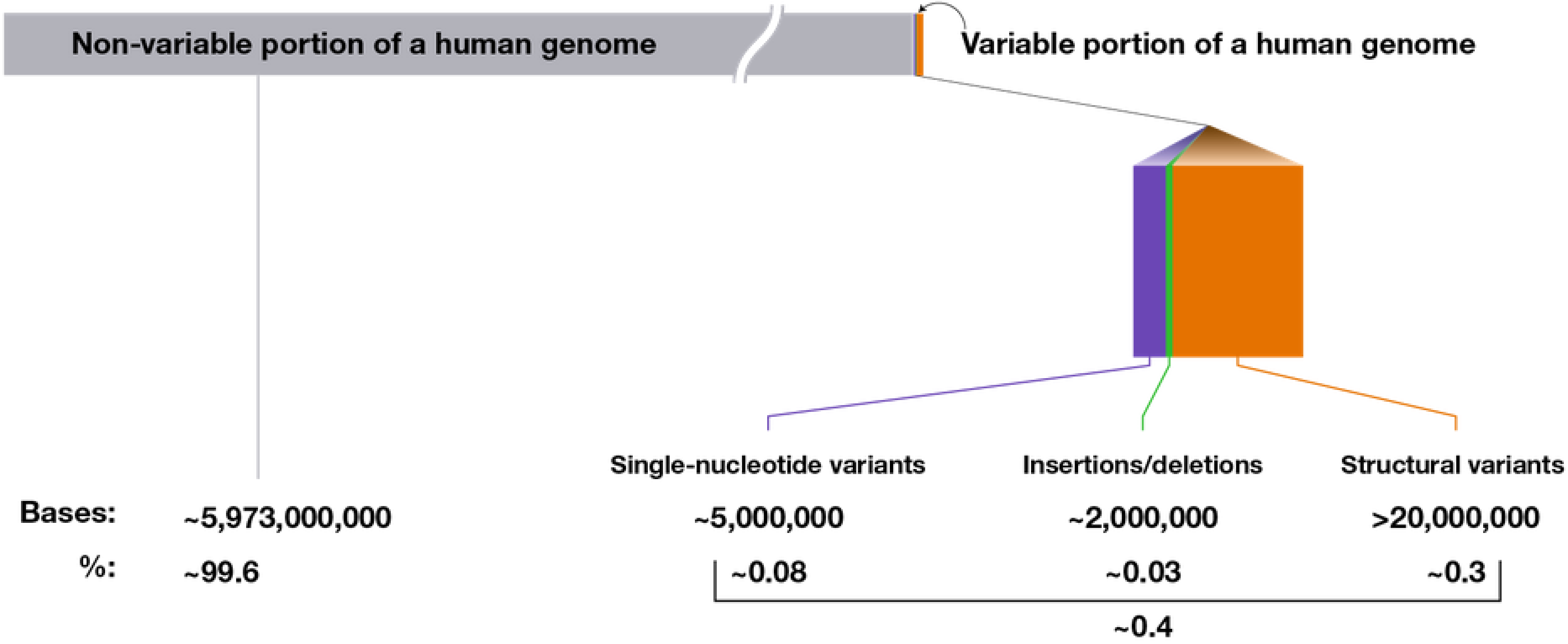
Genomic variation also extends beyond small stretches of nucleotides to larger chromosomal regions. These large-scale genomic differences are called **structural variants** and involve at least 50 nucleotides and as many as thousands of nucleotides that have been inserted, deleted, inverted or moved from one part of the genome to another. Tandem repeats that contain more than 50 nucleotides are considered structural variants; in fact, such large tandem repeats account for nearly half of the structural variants present in human genomes. When a structural variant reflects differences in the total number of nucleotides involved, it is called a copy-number variant (CNV). Note that CNVs are distinguished from other structural variants, such as inversions and translocations, because the latter types often do not involve a difference in the total number of nucleotides.

What Is the Inventory of Genomic Variants in a Typical Human Genome Sequence?



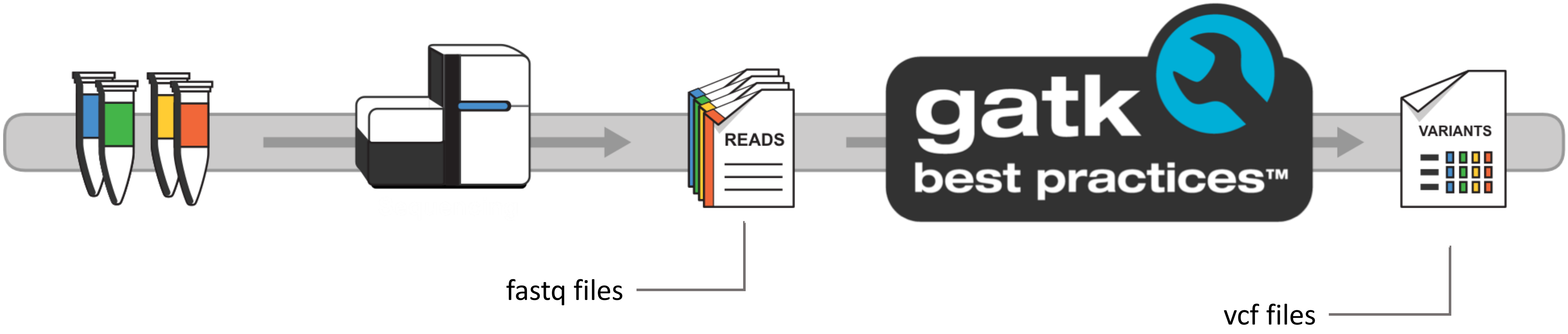
- What does a typical **human genome** sequence look like with respect to the variants that it contains? On average, compared to a reference human genome, a person's ~6 billion-nucleotide genome sequence will have:
- **~5,000,000 SNVs** that involve **~5,000,000 nucleotides**
- **~600,000 insertion/deletion variants** that involve **~2,000,000 nucleotides**
- **~25,000 structural variants** that involve **>20,000,000 nucleotides**.
- These numbers represent current estimated averages and will likely change as more human genomes are sequenced to completion. Furthermore, multiple approaches can be used to calculate how the different types of genomic variants account for the differences among genomes. In fact, calculating the total number of nucleotides involved in structural variants in each **human genome** is quite complicated and remains an active area of research.

What Is the Inventory of Genomic Variants in a Typical Human Genome Sequence?



On average, a **human genome** sequence (consisting of ~6,000,000,000 nucleotides) is ~99.6% identical — or non-variable — compared to a reference human genome sequence. The variable portion of a human genome sequence totals ~0.4%, which consists of the indicated contributions from each of the three major types of genomic variants. **These same general numbers are also found when directly comparing two peoples' genome sequences (as opposed to comparing to a reference human genome sequence).**

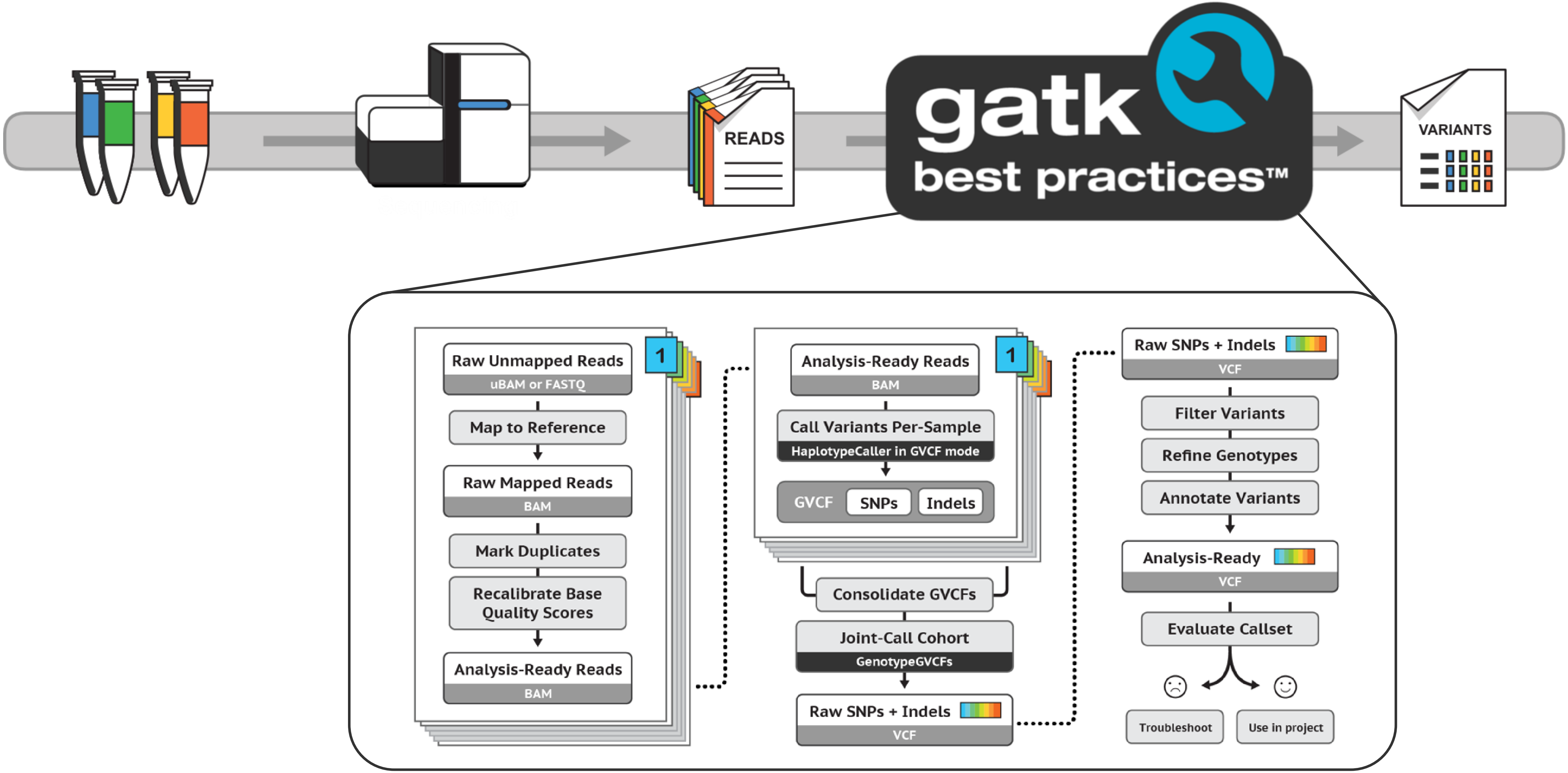
Variant discovery : identify variants in sequencing data



Variant calling pipeline : from raw sequencing reads to VCF

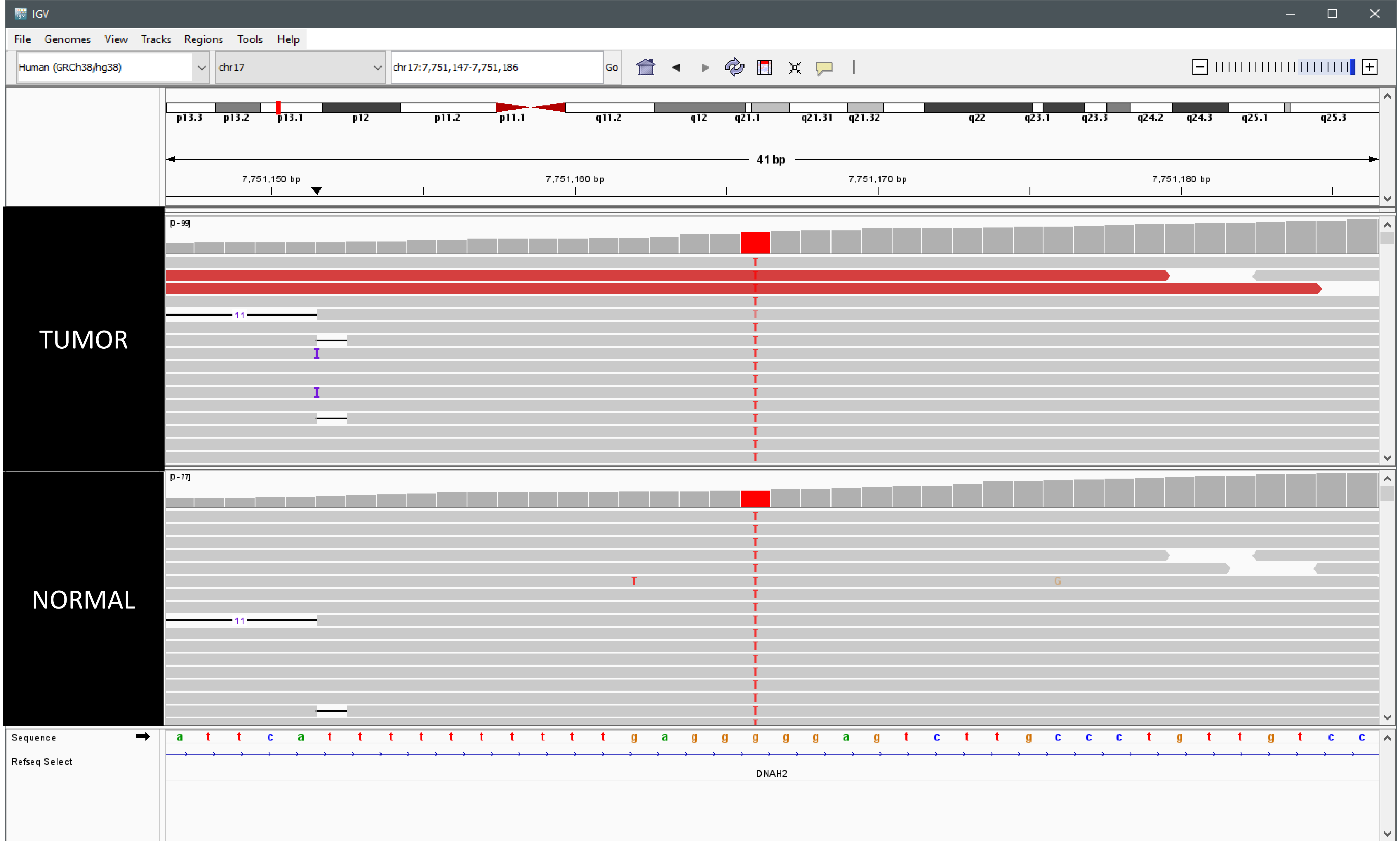
Variant discovery is the process of detecting differences between a sample genome and a reference genome

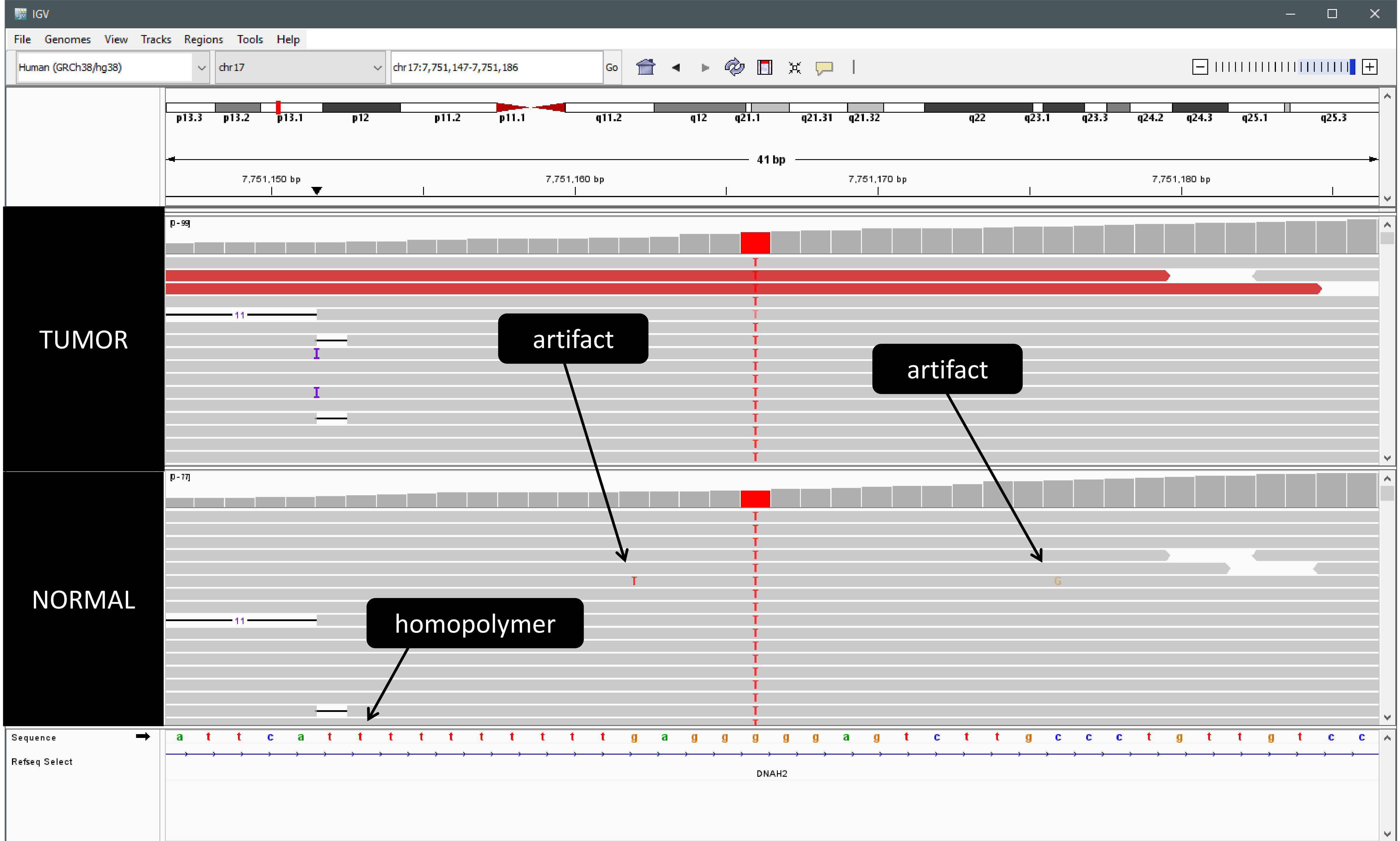
Variant discovery : identify variants in sequencing data



The Challenge of Distinguishing True Variants from Artifacts

A **genetic variant** is a genuine change in the DNA sequence that is actually present in the genome of the individual or a cell population. In contrast, an **artifact** is an *observed* change in the sequence data that isn't real. These are often caused by errors during the sequencing process itself, like a misread base or a problem with the alignment of the sequencing reads to the reference genome.


IGV
File Genomes View Tracks Regions Tools Help
Human (GRCh38/hg38)
chr17
chr17:7,751,147-7,751,186
Go
p13.3 p13.2 p13.1 p12 p11.2 p11.1 q11.2 q12 q21.1 q21.31 q21.32 q22 q23.1 q23.3 q24.2 q24.3 q25.1 q25.3
41 bp
7,751,150 bp
7,751,160 bp
7,751,170 bp
7,751,180 bp
TUMOR
NORMAL
Sequence
Refseq Select
DNAH2


IGV
File
Genomes
View
Tracks
Regions
Tools
Help
Human (GRCh38/hg38)
chr17
chr17:7,751,147-7,751,186
Go
p13.3
p13.2
p13.1
p12
p11.2
p11.1
q11.2
q12
q21.1
q21.31
q21.32
q22
q23.1
q23.3
q24.2
q24.3
q25.1
q25.3
41 bp
7,751,150 bp
7,751,160 bp
7,751,170 bp
7,751,180 bp
TUMOR
artifact
artifact
NORMAL
homopolymer
Sequence
Refseq Select
DNAH2

Common Next-Generation Sequencing Artifacts

Sequencing-Specific Artifacts

These are errors introduced during the DNA sequencing process itself. They are often platform-specific and can appear as "variant calls" at positions that aren't actually different from the reference.

- ***Low-Quality Bases:*** *A base call with a low Phred score (e.g., Q < 30) has a higher chance of being a misread. These can look like a variant, but are simply an error.*
- ***Read-Through or Contamination:*** *If DNA libraries were not properly cleaned, adapter sequences can be read and incorrectly aligned, leading to false variants.*
- ***Strand Bias:*** *This occurs when a potential variant is almost exclusively observed on reads mapped to only one strand (either the forward or reverse strand). A* **true variant** *should be equally represented on both strands. If it appears only on one, it often indicates a technical error, such as damage to the DNA during library preparation or a systematic error during the sequencing cycle, leading to a high-confidence false positive.*

Alignment-Specific Artifacts

These occur when the sequencing reads are aligned to the reference genome, even if the reads themselves are high-quality.

- ***Misalignment near Indels:*** *Reads containing a small insertion or deletion (indel) can be difficult to align correctly. They might be shifted slightly, causing nearby bases to look like false single nucleotide variants.*
- ***Reads Misaligned to a Paralogue or Pseudogene:*** *A read from a gene might incorrectly align to a similar-looking pseudogene (a non-functional copy of the gene) somewhere else in the genome. This makes the entire read look "different" and can cause a flood of false variant calls.*

Low Coverage and Allele Drop-out

This is a broad category covering issues related to insufficient or uneven read depth, which can compromise variant detection.

- ***Low Depth/Coverage:*** *If a position is covered by very few reads (e.g., <5), the statistical power to distinguish a true variant from a random sequencing error is extremely low.*
- ***Allele Drop-out (ADO):*** *In heterozygous regions (Ref/Alt), one of the two alleles is sometimes preferentially amplified or sequenced poorly. If the Alt allele drops out, the position is falsely called Homozygous Reference (Ref/Ref). Conversely, if the Ref allele drops out, the position is falsely called Homozygous Alternate (Alt/Alt).*

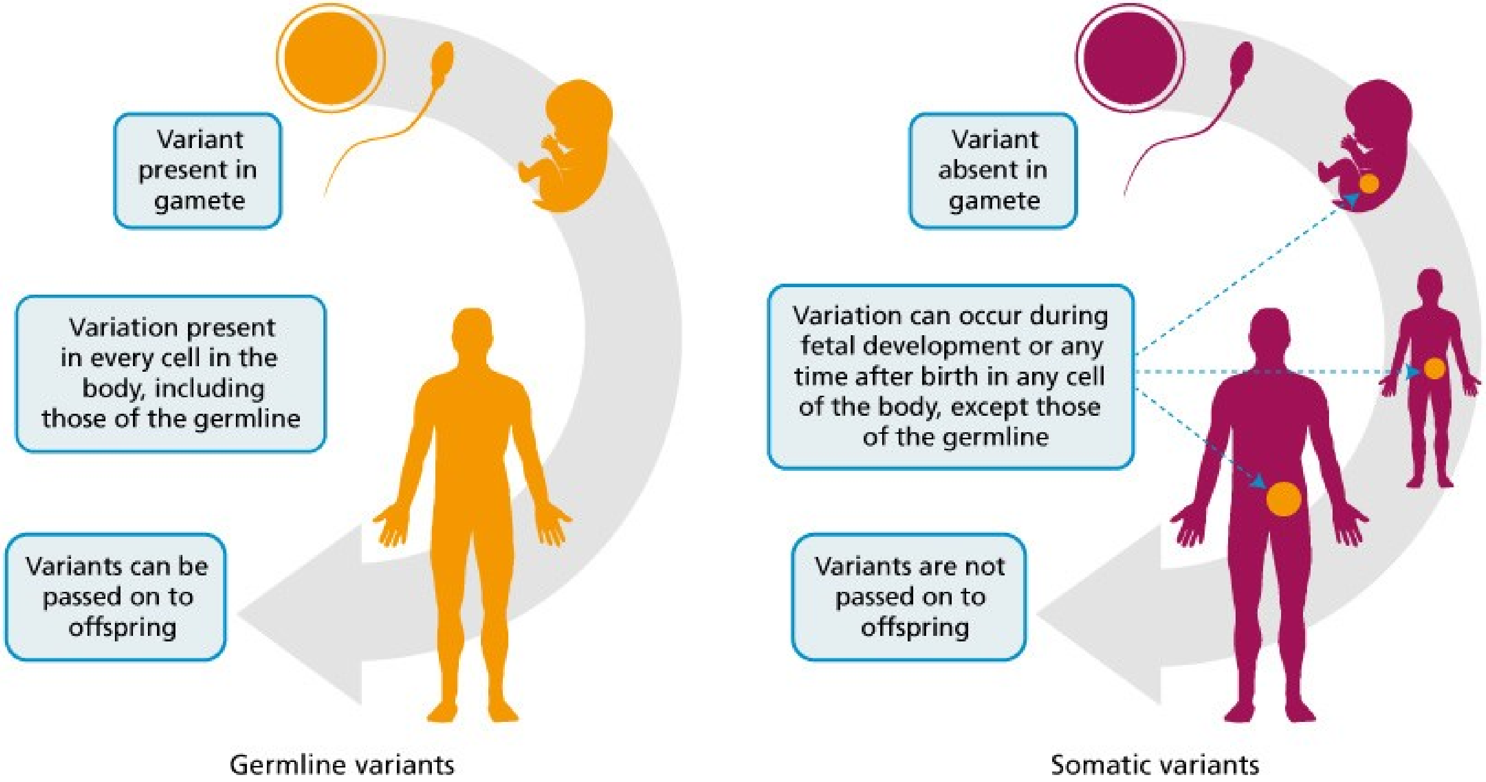
Understanding the Landscape of Genetic Variation
Germline vs. Somatic Variants

- **Germline Variants**

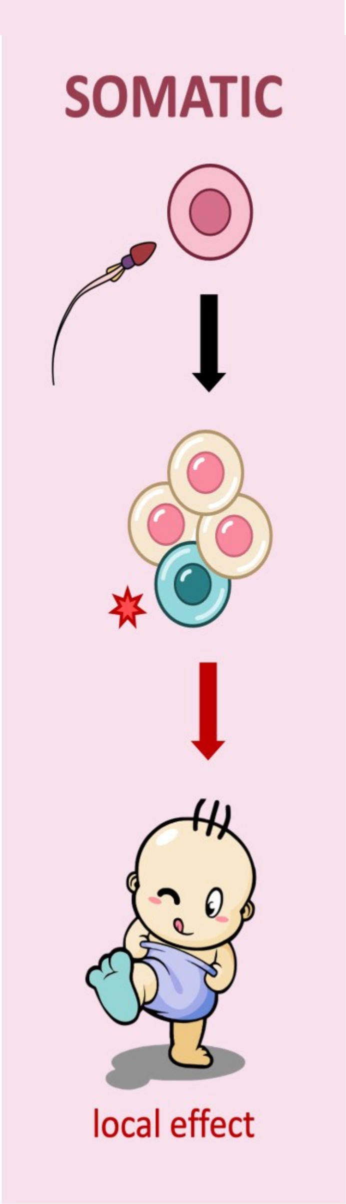
  These are inherited from a parent and are present in every cell of an individual's body. They arise from genetic mutations in the egg or sperm cells. Calling germline variants is a key component of understanding inherited disease risk, population genetics, and human evolution. For example, a single nucleotide polymorphism (SNP) associated with a higher risk for a certain condition would be a germline variant.

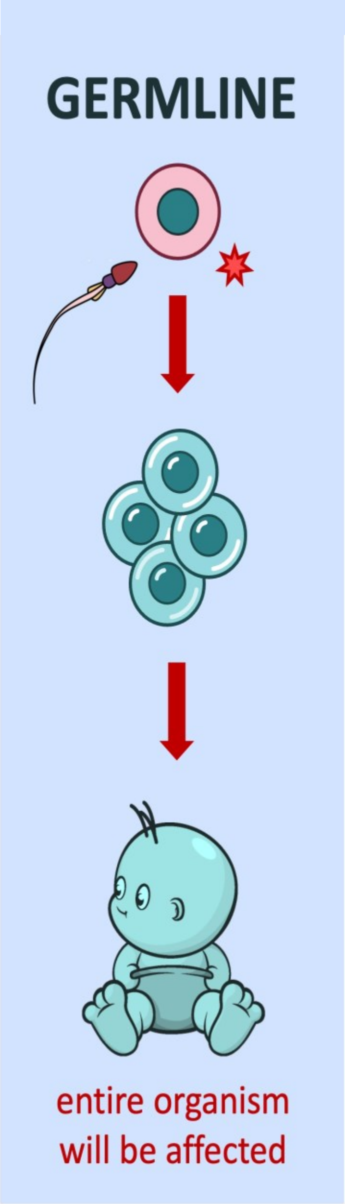
- **Somatic Variants**

  These are non-inherited genetic changes that occur after conception. They are not found in every cell of the body but are present in specific cell populations, such as a tumor. A classic example is a cancer-driving mutation like a change in the EGFR gene, which is found only in the tumor cells and not in the surrounding healthy tissue. Identifying somatic variants is the cornerstone of cancer genomics and targeted therapy.



Somatic mutations vs germline mutations



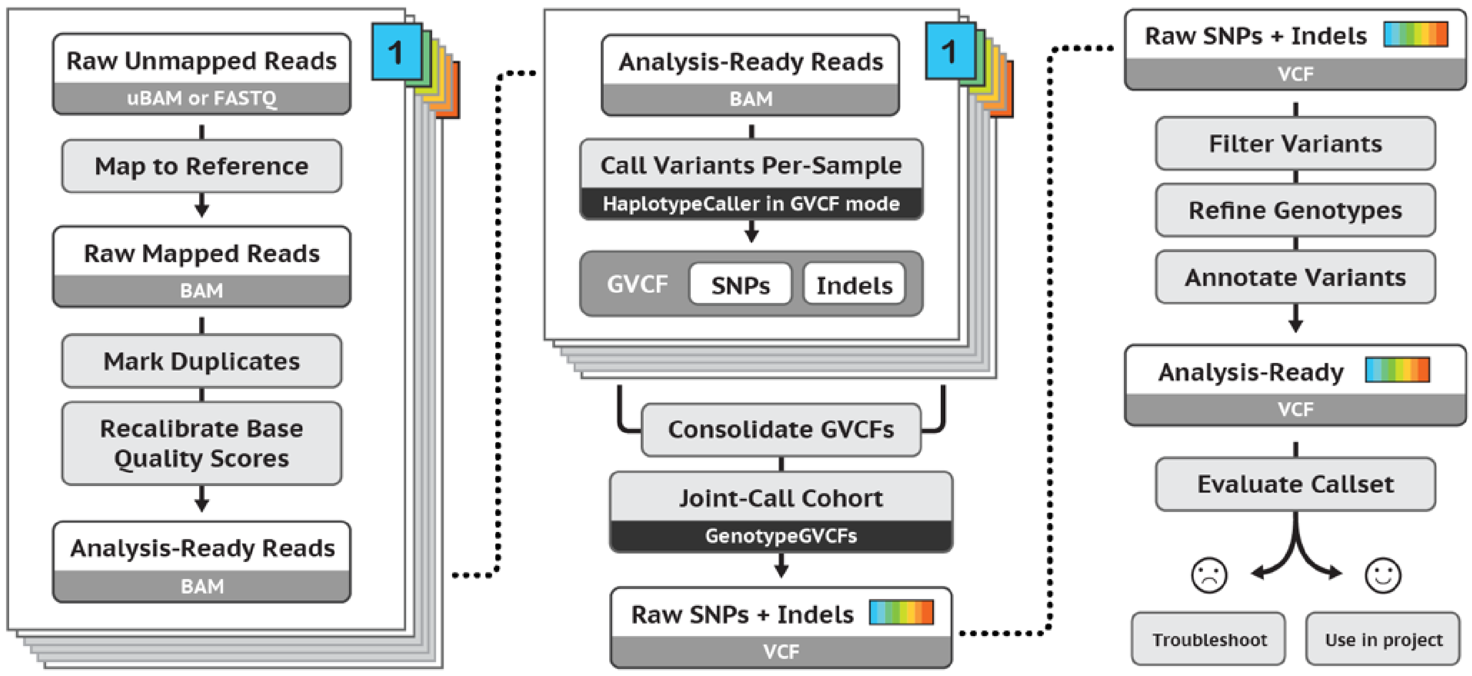


Somatic mutation	Germline mutation
It is defined as any alteration in the genetic sequence of genes of the somatic cells.	It is also called “acquired mutation” as it is acquired during an individual’s life.
It is defined as any alteration in the genetic sequence of genes of the germinal cells.	It is also called “hereditary mutation” as it is passed to offspring.
It occurs in somatic or body cells.	It occurs in germ cells.
It is not inheritable.	It is inheritable.
It affects only mutated cells and their progeny.	It affects all the cells of the organism.
It can occur at any stage of the life cycle.	It occurs only during gametogenesis.
It plays no role in evolution.	It is the basis of evolution.
It does not cause genetic disorders but may cause cancer.	It is responsible for genetic disorders and also germline cancers.
It is finished along with the death of the individual.	It continues till the offspring of the individual breed and may result in separate sub-species. Hence it is genetically more important.
In most cases, they show observable effects.	In most cases, they are silent and don’t show detectable effects.
They can be treated or cured.	They can’t be treated or cured.
Mosaicism occurs.	Usually, mosaicism doesn’t occur.
Examples include cancers, tumor development, a neurodegenerative disorder, etc.	Examples include Hemophilia, Down’s Syndrome, 18-Trisomy, etc.

https://s3-us-west-2.amazonaws.com

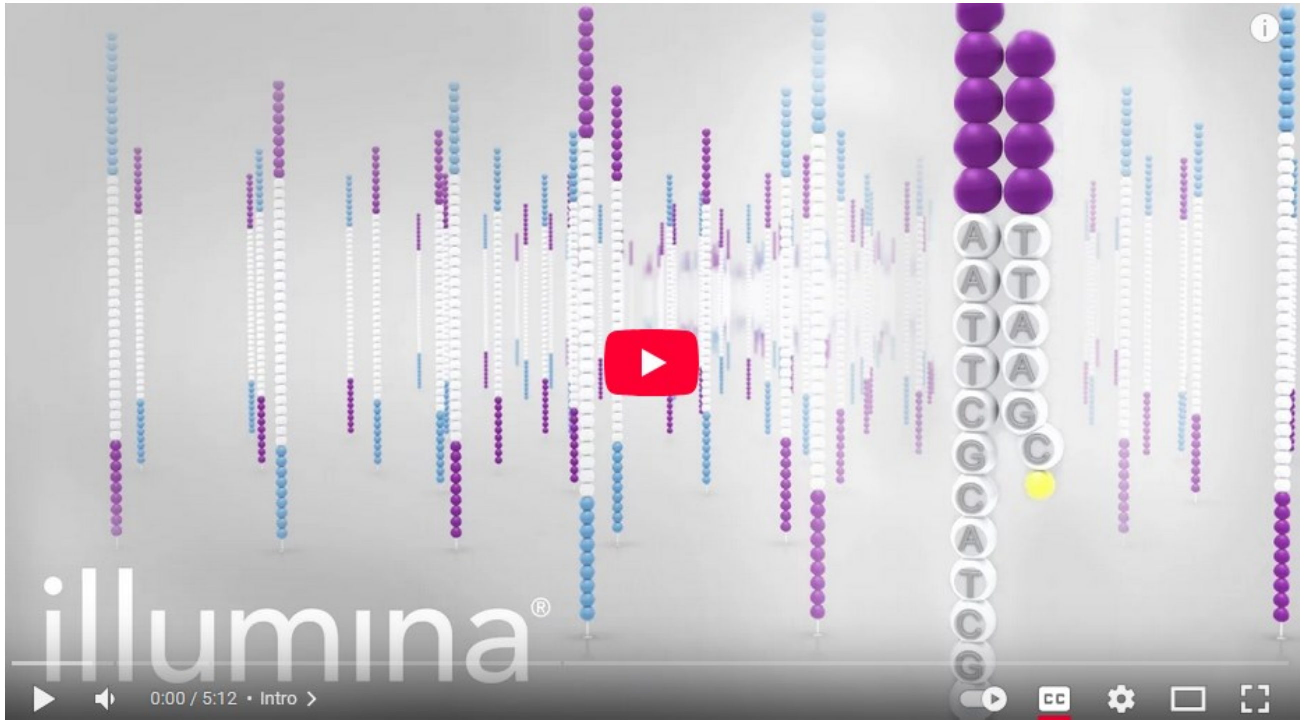
SECTION BREAK

Phases of variant calling



1. Sequence data processing
2. Variant calling
3. Variant filtering
4. Variant annotation
5. Evaluation & QC

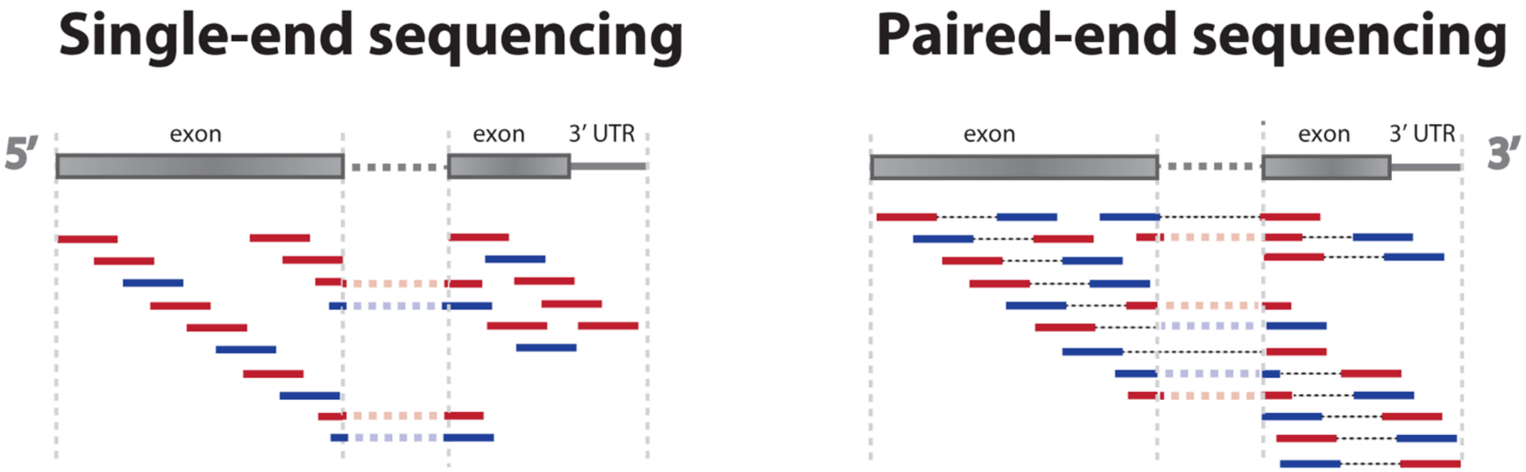
Sequencing by synthesis

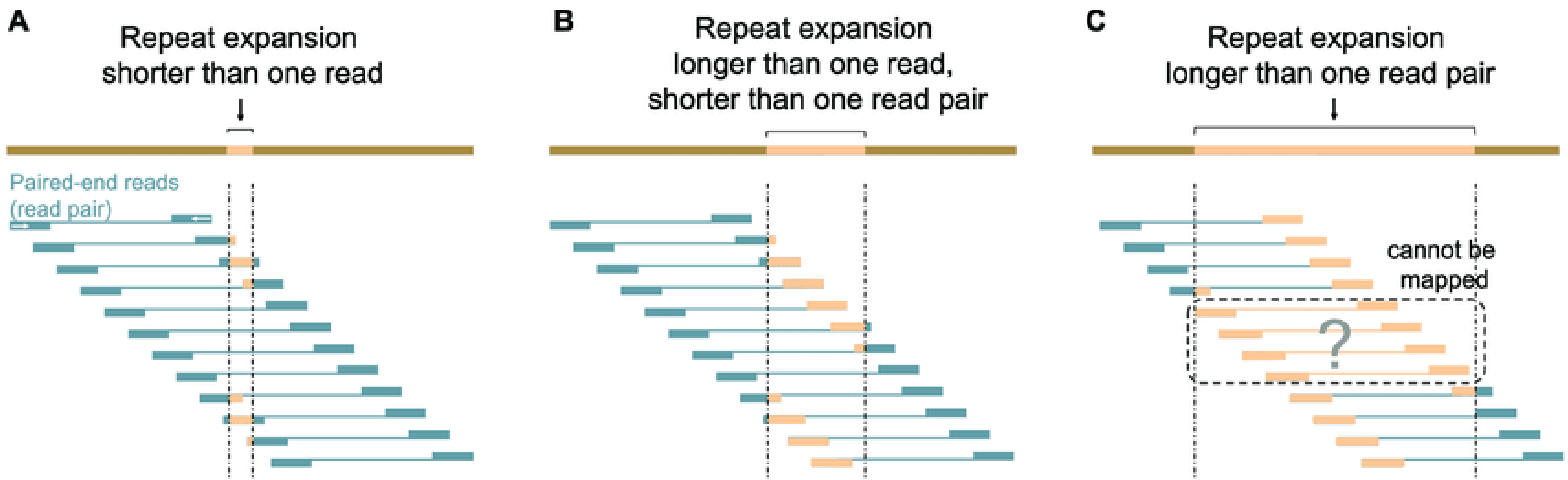


https://www.youtube.com/watch?v=fCd6B5HRaZ8

Overview of Illumina Sequencing by Synthesis

Single-end/Paired-end sequencing

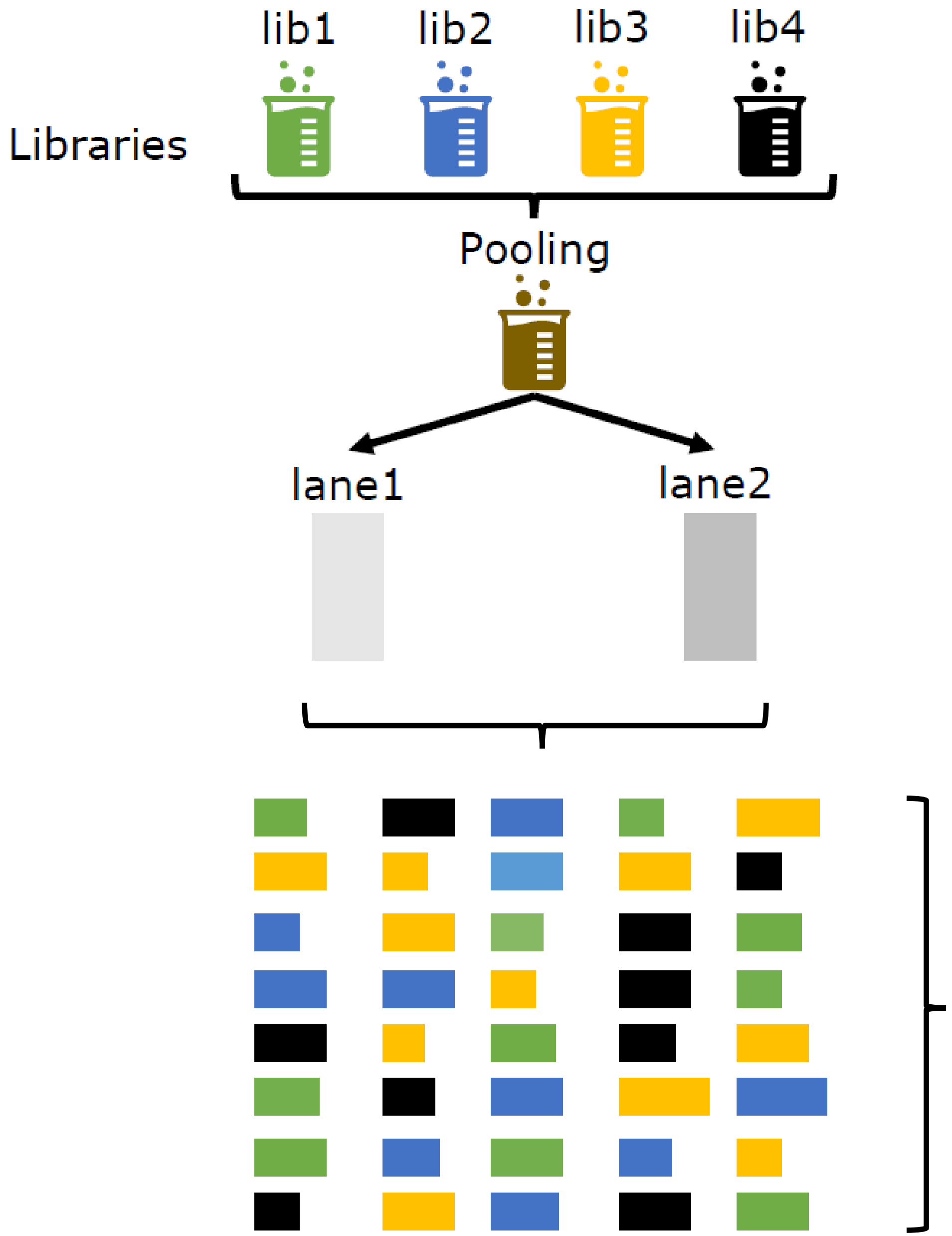




Paired-end gives us more information, especially useful in detecting indels or structural changes.

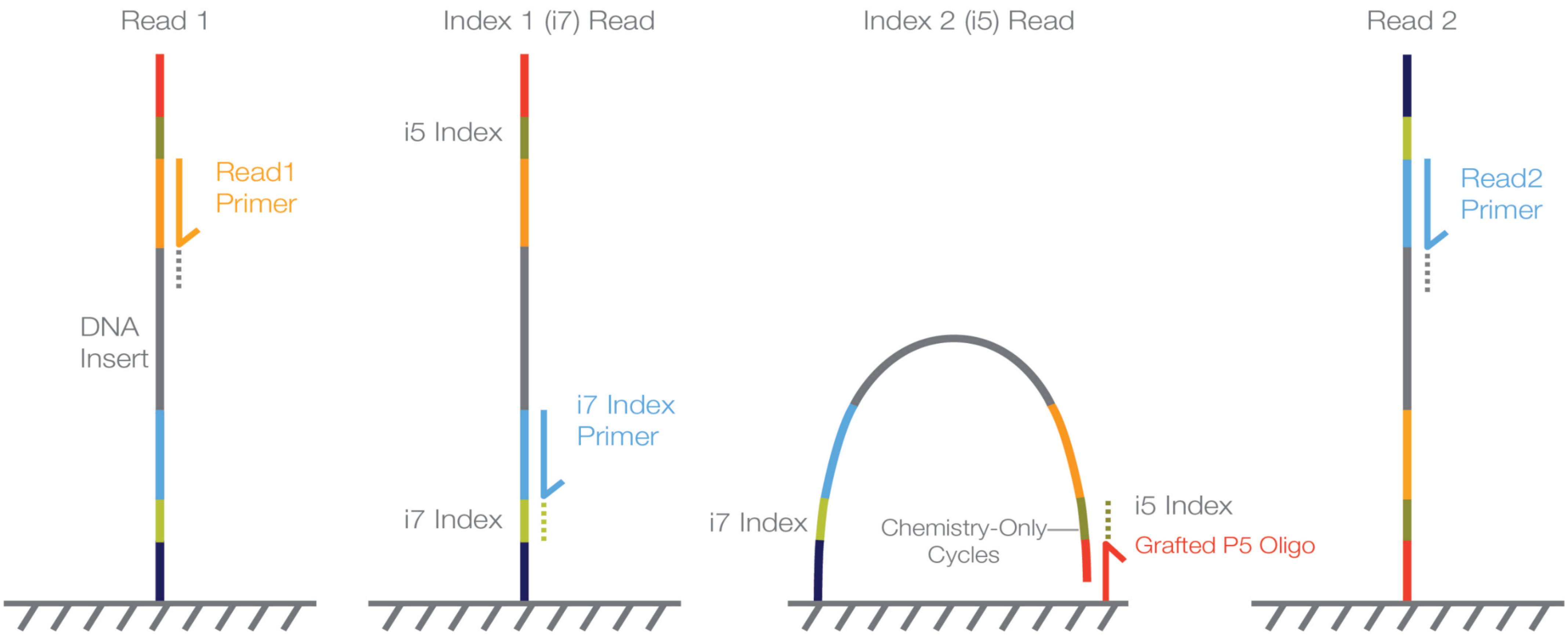
DOI:[10.1016/j.parkreldis.2020.02.015](https://doi.org/10.1016/j.parkreldis.2020.02.015)

Multiplexing/Demultiplexing



Pooling (Multiplexing)
Combining multiple DNA libraries (from different samples) into a **single sequencing run.**
Each sample is prepared with a **unique index (barcode)** embedded in the adapters (usually in the P7/P5 adapters).

Demultiplexing (splitting the reads)
Sorting reads from the pooled FASTQ into separate files **based on the index sequences**.



Adding unique barcodes to the index reads allows different samples to be sequenced concurrently. Following this, the unique index reads are used to separate out the original samples.

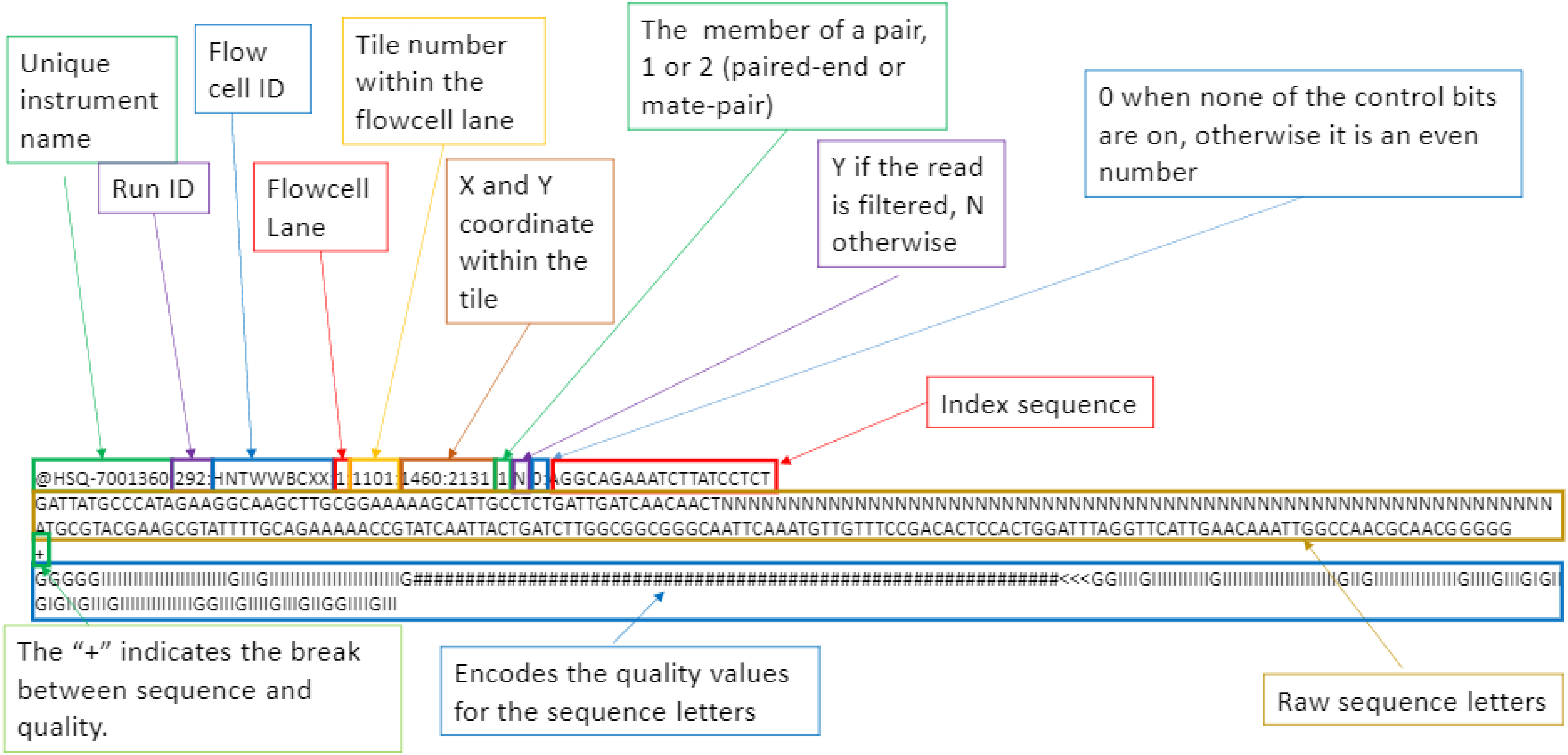
https://support.illumina.com/documentation.html

Raw Sequencing Reads (FASTQ)

identifier
sequence
+
quality
identifier
sequence
+
quality

```
@M00877:474:000000000-K7HDK:1:1101:10371:2108 1:N:0:TCTCATGA+CTTCTAAT
TTTATGGCCTACGGGAGGCAGCAGTGAGGAATATTGGACAATGGAGGCAACTCTGATCCTGCCATTCCGCGTGTTTGTCTACGTCCCTATGTTCTTTTTTCTTCTTTTTTCTCTG
+
-ACCCGGGGGGGGGGGGGGGGGGGGCFGFGF9FFADEF8EFE<A,,8,,:CFF,,;6;,,;;,;,,<,6+8+,,,,,,:,,,,,66:,:,,,,<,9,,+495,CCE,,+4,,,,
@M00877:474:000000000-K7HDK:1:1101:10833:2162 1:N:0:TCTCATGA+CTTCTAAT
TATTAAGCCTACGGGCGGCAGCAGTGGGGAATATTGCACAATGGGCGCAAGCCTGATGCTGCCATGCCGCGTGTATGAAGAAGGCCTTCTGTTTTTTAAGTACTTTCTTCTTTGA
+
@ACCCGGGGGGGGGDGGGGGGGGGGGGGGGGGGGGGGGGGGGEC@+@76DF8,C,6,,C6C?,6C+7+8+B,C,,,,,,,,,9:?E,,,C?,D+,,,:,:CEF,,,,,,,,,
@M00877:474:000000000-K7HDK:1:1101:16190:2217 1:N:0:CCTCATGA+CGTCTAAT
CGCTAGGCCTACGGGAGGCAGCAGTGGGGAATATTGCACAATGGGCGAAAGCCTGATGCAGCAACGCCGCGTTAGCGATGAAGGCCTTCTGTTCGTTAAGCTCTTTCCTCAAGGA
+
@BCCCGGGGGGGGGGGGGGGGGFGGGGGGGGGGGGGGGGGGD77+@C<FGD8CC<,9C,,:++@+@+8+,,,4+:,,,,444C=F,,,C9+B,++,5AED,B55AA,,,,,
@M00877:474:000000000-K7HDK:1:1101:20697:2220 1:N:0:GCTCATGA+CGTCTAAT
TGACCTACCTACGGGAGGCTGCAGTGAGGAATATTGGTCAATGGTCGCGAGACTGATCCTGCCCTGTAGCGTTCATGTCGACTGCCCTCCGTTTTTTTTTCTGCTTTTTTTTTTG
+
```

Raw Sequencing Reads (FASTQ)

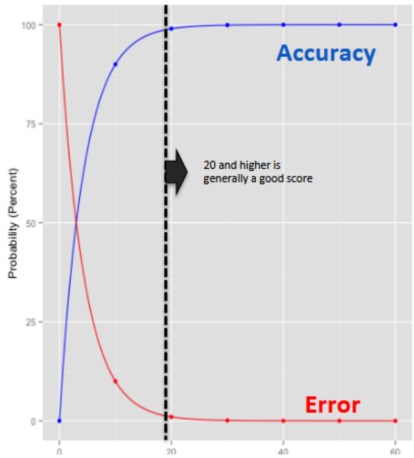


Quality (Phred) scores

Phred value = -10*log10(err)

Example:

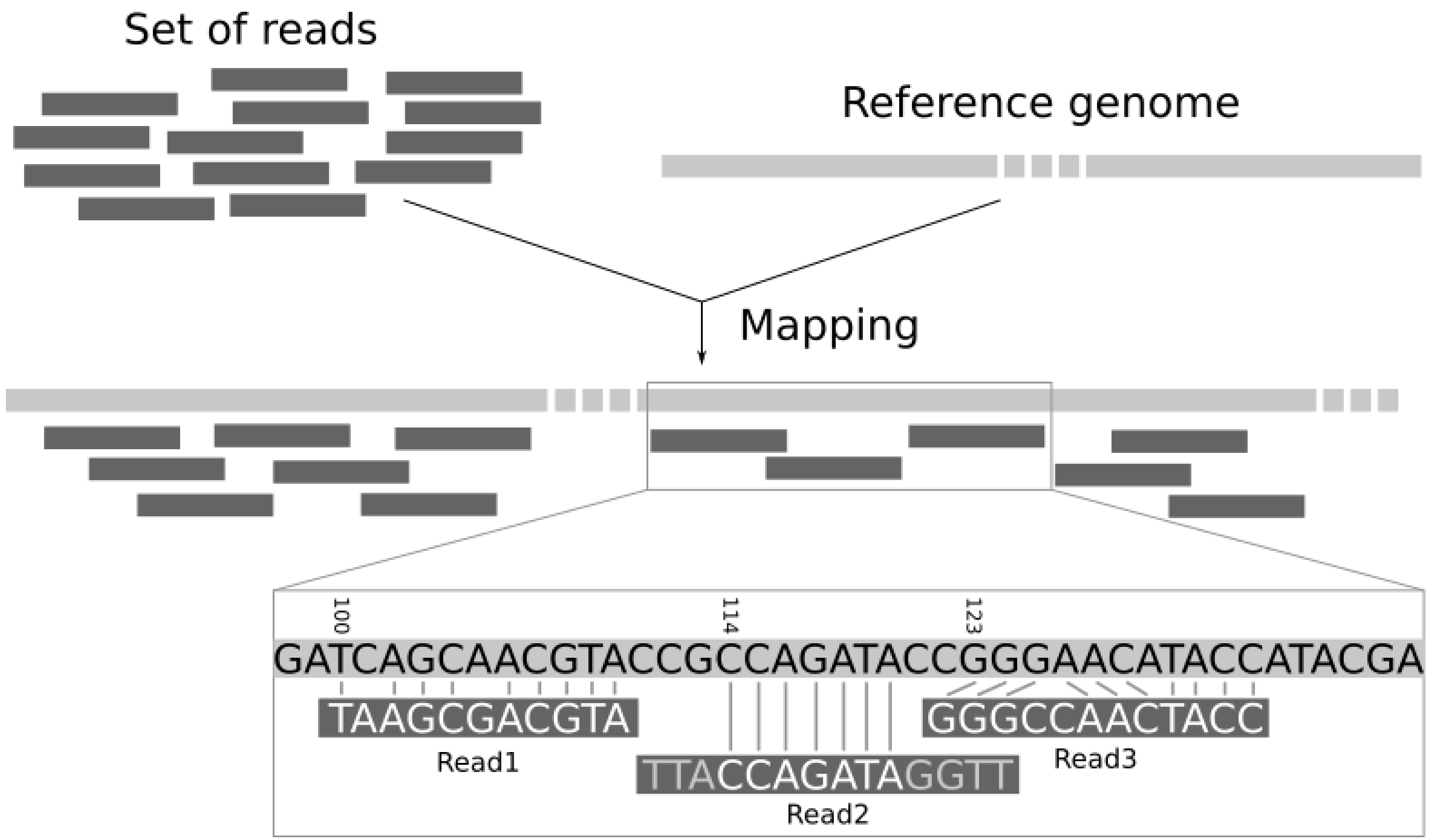
- 90.0% confidence (10% error rate) = Q10
- 99.0% confidence (1% error rate) = Q20
- 99.9% confidence (0.1% error rate) = Q30



Illumina v1.8 and later (ASCII_BASE=33)

Q	ASCII	P	Q	ASCII	P	Q	ASCII	P	Q	ASCII	P
1	"	0.79433	12	-	0.06310	23	8	0.00501	34	C	0.00040
2	#	0.63096	13	.	0.05012	24	9	0.00398	35	D	0.00032
3	$	0.50119	14	/	0.03981	25	:	0.00316	36	E	0.00025
4	%	0.39811	15	0	0.03162	26	;	0.00251	37	F	0.00020
5	&	0.31623	16	1	0.02512	27	<	0.00200	38	G	0.00016
6	'	0.25119	17	2	0.01995	28	=	0.00158	39	H	0.00013
7	(	0.19953	18	3	0.01585	29	>	0.00126	40	I	0.00010
8	)	0.15849	19	4	0.01259	30	?	0.00100	41	J	0.00008
9	*	0.12589	20	5	0.01000	31	@	0.00079			
10	+	0.10000	21	6	0.00794	32	A	0.00063			
11	,	0.07943	22	7	0.00631	33	B	0.00050			

2. Alignment to Reference Genome (BAM/CRAM)



- Align short reads to a known reference (e.g., GRCh38).
- Tools: BWA-MEM, Bowtie2, minimap2.
- Output: SAM/BAM/CRAM file (binary alignment).
- Why? To know where each read comes from in the genome.

Challenges:

- Reads mapping equally well to multiple locations.
- Indels near repetitive regions.

```
  131       141       151       161       171       181       191
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAAGCGAACCACAGTCAAAACGCCCTCCTGGC  ← reference
..........................................................................
CATC           CCTCTCCCGCACCTTCGGATATACTATCAAGCGAACCACAGTCAAAACGC    CTGGC
CATCTT             TCCCGCACCTTCGGATATACTATCAAGCGAACCACAGTCAAAACGCCCTC    C
CATCTTCAACACCCGCC   CCCGCACCTTCGGATATACTATCAAGCGAACCACAGTCAAAACGCCCTCC
CATCTTCAACACCCGCCTCT   GCACCTTCGGATATACTATCAAGCGAACCACAGTCAAAACGCCCTCCTGG
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATAT    TCAAGCGAACCACAGTCAAAACGCCCTCCTGGC
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTA     CGAACCACAGTCAAAACGCCCTCCTGGC
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAA    ACCACAGTCAAAACGCCCTCCTGGC
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAAGCGAA  ACAGTCAAAACGCCCTCCTGGC
CATCTTCAA                                              GTCAAAACGCCCTCCTGGC
CATCTTCAACACC                                                   CCCTCCTGGC
CATCTTCAACACCCGCCTCTC
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAAGCGAA
  TCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAAGCGAACC
   CTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAAGCGAACCA  ← read
```

SAM/BAM file format

@RG: Read group
ID: Read group identifier
LB: Library
PL: Platform
SM: Sample

Reference sequence name

Header line

Sorting order of alignments: coordinate

Reference sequence length

Example BAM file

Header

```
@HD VN:1.5  SO:coordinate
@SQ SN:chr1 LN:248956422
@SQ SN:chr2 LN:242193529
...
@RG ID:Case_003_T   LB:Case_003_T   PL:ILLUMINA     SM:Case_003_T
@PG ID:bwa  PN:bwa  VN:0.7.17-r1188 CL:bwa mem -t 10 -M -R ...
```

Reads

```
Read1 99 chr1 10147 40 10M    = 10395 310 ACCCTAACCC CHJ<GGIF;G RG:Z:Case_003_T
Read2 83 chr1 10178 22 7M6H3M = 10201 -65 CCCTATCCCG F=(2B9DB9) RG:Z:Case_003_T
```

Read name

Position

Mapping quality

Sequence

Base quality

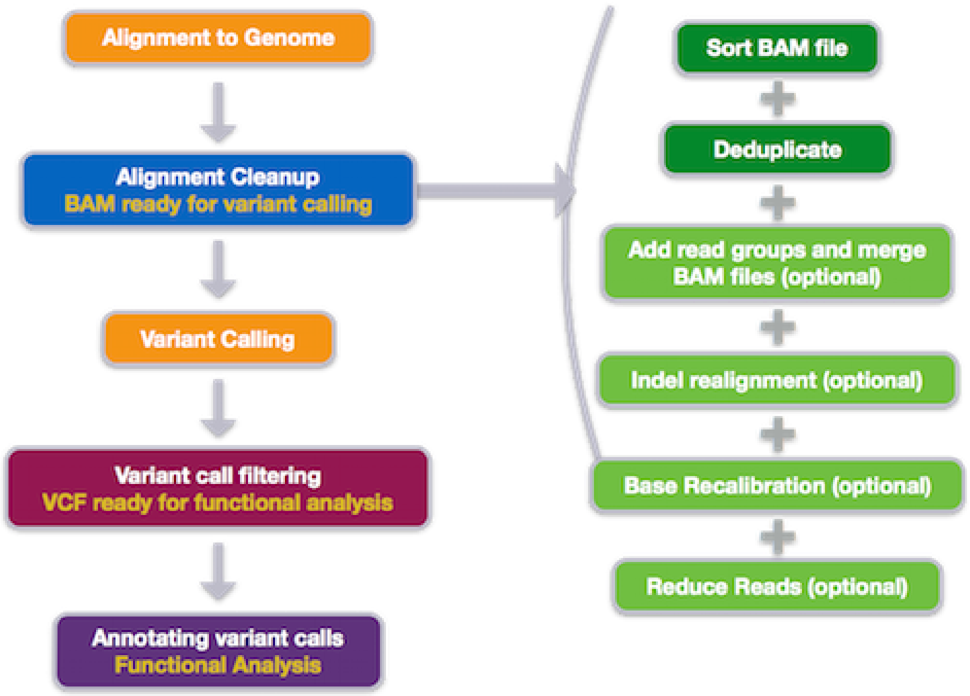
@PG Program
ID: Program record identifier
PN: Program name
VN: Program version
CL: Command line.

Base quality scores are per-base estimates of error emitted by the sequencing machines.
Short variant calling algorithms rely heavily on base quality scores

Preprocessing

Before calling variants, BAM files are *cleaned*:

- **Sorting**: order reads by coordinate.
- **Marking duplicates**: PCR duplicates can bias allele counts.
- **Base Quality Score Recalibration** (BQSR): adjust quality scores based on systematic errors.



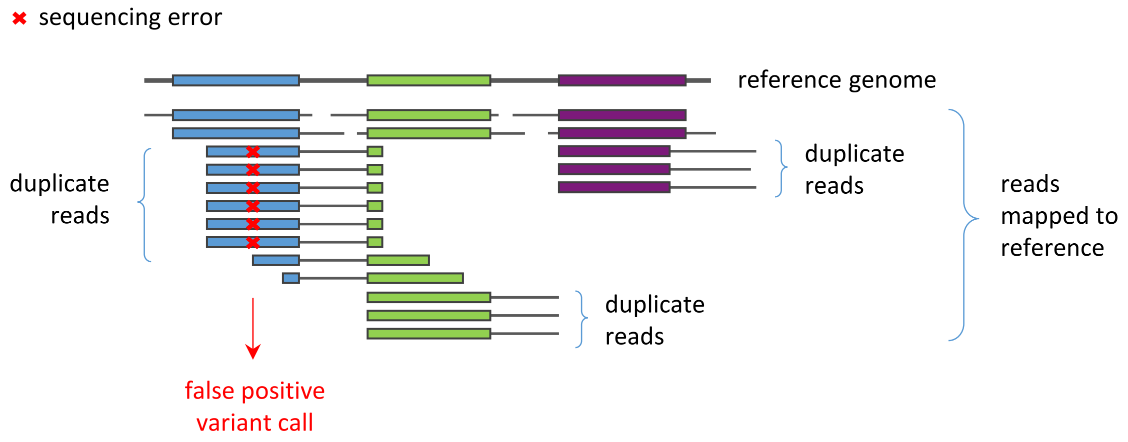
Marking/removing duplicates

Duplicate reads = multiple reads from the same original DNA fragment

- **PCR duplicates**: created during library amplification
- **Optical duplicates**: same cluster mistakenly counted more than once

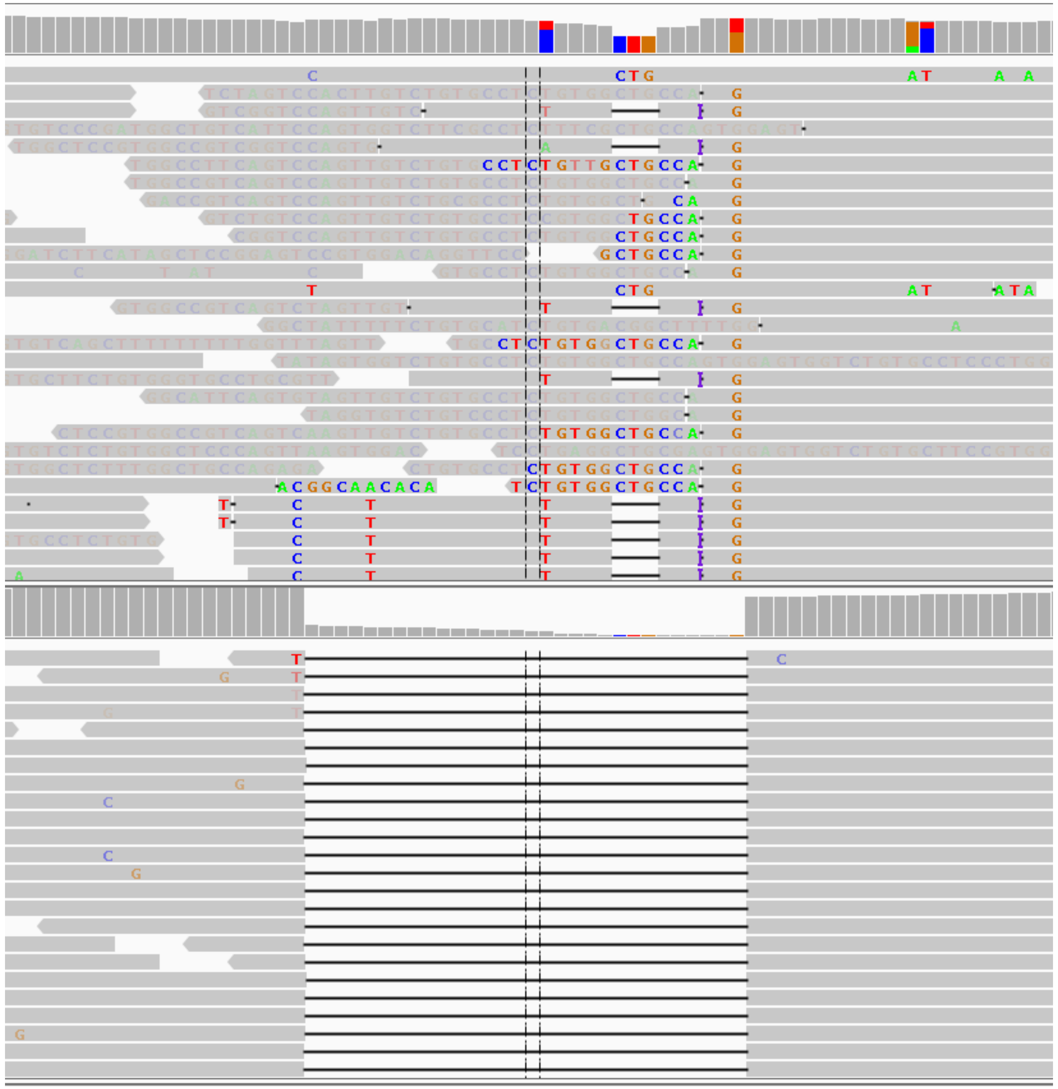
Why remove them?

- Variant calling assumes each read is an independent observation
- Duplicates share the same origin → not independent evidence



After flagging duplicates, it is more likely that the right call will be made

Indel realignment



Before indel realignment

misaligned reads around insertions/deletions → false variant calls

After indel realignment

reads are properly aligned → more accurate variant calling

https://raw.githubusercontent.com/mozack/abra/master/misc/example.png

Base quality score correction

- Each base in a sequencing read has a **quality score (Phred score)**
- Reflects the **probability that the base was called incorrectly**
- Essential for estimating genotype likelihood

Sequencers can make **systematic errors** depending on:

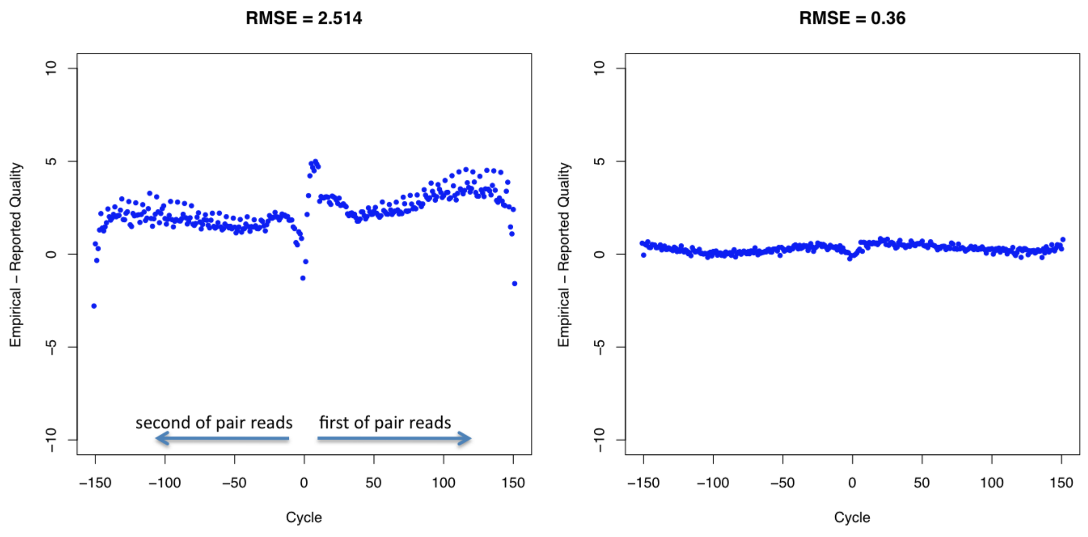
  - **Cycle number**: the quality may drop toward the end of the read.
  - **Base context**: some sequences (like “GGC”) are more error-prone.
  - **Machine-specific biases**: differences between flow cells, runs, or lanes.

Base Quality Score Recalibration (BQSR)

Base Quality Score Recalibration (BQSR)

- Adjusts sequencer-reported base quality scores to match real error rates
- Corrects for systematic biases: read cycle, sequence context, machine-specific effects
- Makes variant calling more accurate by trusting high-quality bases and down-weighting error-prone ones

Residual Error by Machine Cycle

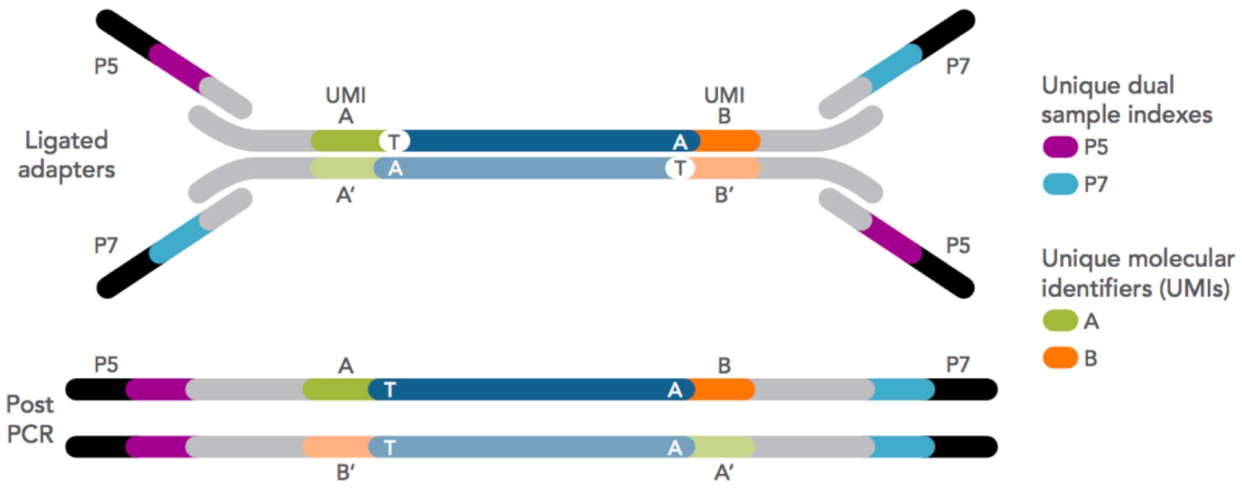


Original data

After recalibration

https://gatk.broadinstitute.org

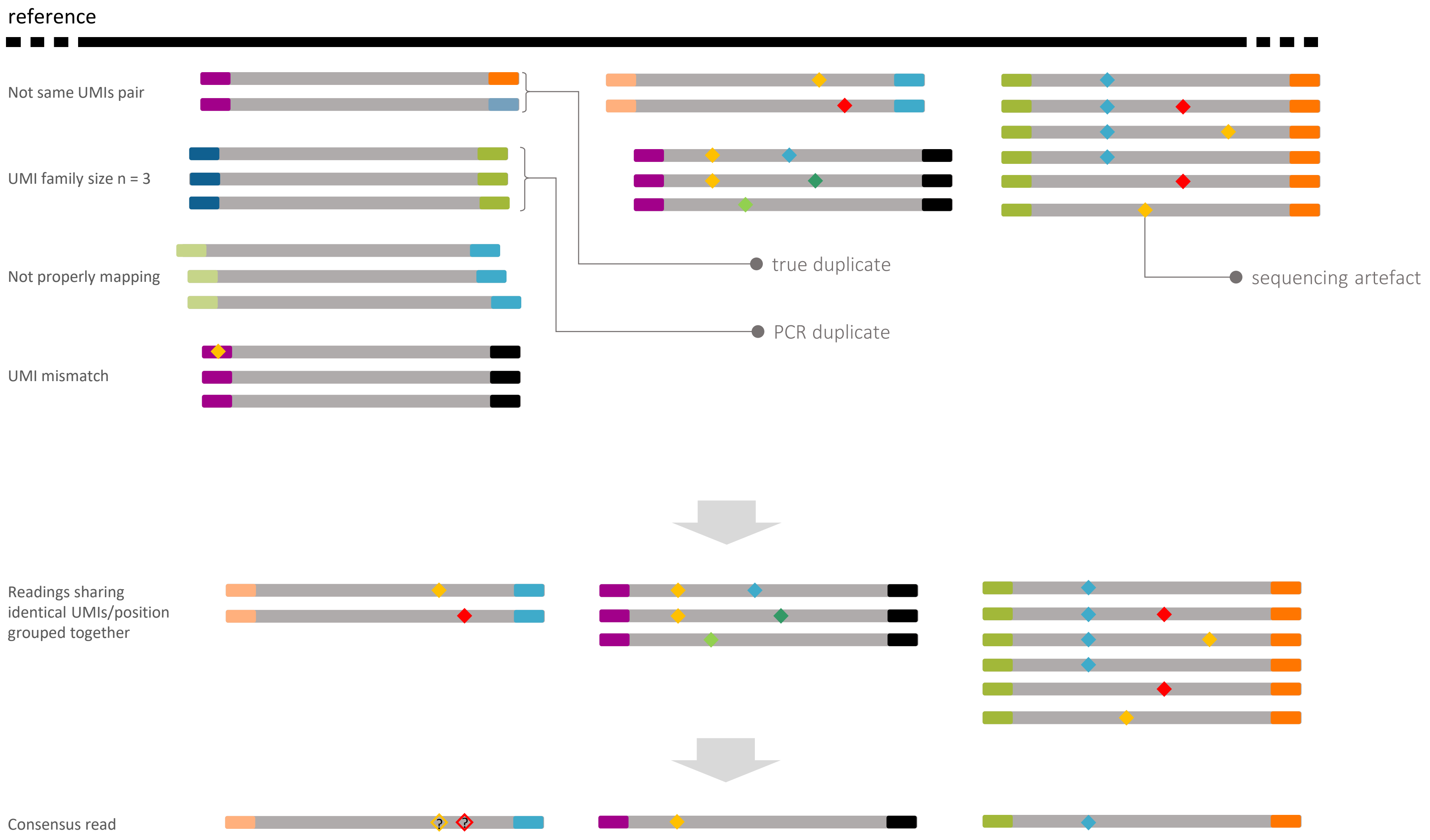
UMIs (Unique Molecular Identifiers)



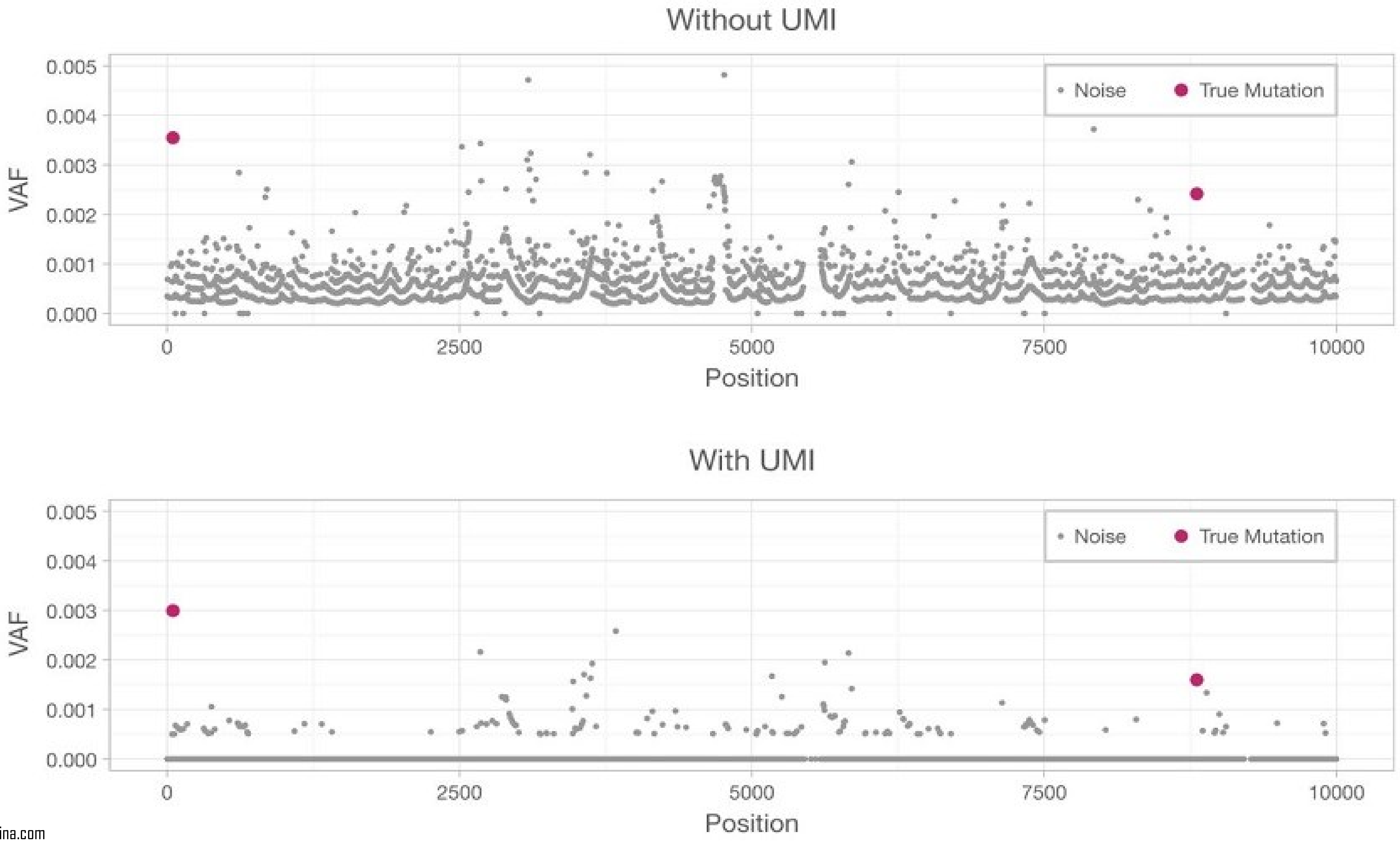
UMIs – Unique Molecular Identifiers

- Short, random sequences added to each DNA/RNA molecule before amplification
- Allow true duplicate identification and consensus calling
- Enable accurate detection of low-frequency variants (e.g., cancer, liquid biopsy)

Consensus read / error correction



Umi pipeline



Illumina.com

4. Variant Calling

Questions

1. Is there a variant at location X?
   Deviation from REF in the alignments

2. What are the alleles?
   The variation in sequence in these deviations

3. What is the genotype (HomRef, Heterozygote or HomAlt)?
   Estimating the allele counts in the sample

4. What is the allele frequency?
   Relative proportion of the alt allele

The Statistical Nature of Variant Calling

Variant calling is fundamentally a statistical process, not a simple pattern match. It's not enough to find a mismatch between the sequenced DNA and the reference genome; the challenge lies in evaluating the evidence for that mismatch and deciding whether it represents a genuine variant or is merely a sequencing artifact. This probabilistic approach is critical because all sequencing technologies have an inherent error rate.

The process relies on three primary sources of evidence to distinguish signal from noise:

- **Sequencing Depth (Coverage)**

  This refers to the number of times a specific position in the genome has been sequenced. It's a key measure of data quality. **A higher coverage provides stronger statistical evidence for a variant**. For example, if you see a mismatch at a position that has been sequenced only twice, the confidence is low. If that same mismatch appears in 100 out of 100 reads, the confidence is exceptionally high.

- **Base Quality Scores**

  Each base call from a sequencer comes with a **Phred score**, a **logarithmically scaled measure of the confidence** in that specific base call. A score of Q30 means there's a 1 in 1,000 chance of the base being wrong. A variant call that is supported by reads with high base quality scores is much more trustworthy than one supported by reads with low-quality scores.

- **Allele Frequency**

  This is the **proportion of sequencing reads that show the variant allele at a given position**. In a standard germline sample, this value is expected to be close to 0% (reference allele), 50% (heterozygous variant), or 100% (homozygous variant). In a tumor sample, however, the allele frequency can be much lower due to the **presence of normal cells** and **sub-clonal populations of cancer cells**.

Germline Variant Calling

In **germline variant calling**, the primary goal is to identify a change that is **present in all**, or nearly all, of the sequenced cells. The expectation is that the variant will be present in roughly **50%** of the reads for a **heterozygous variant** (one copy of the allele) or **100%** of the reads for a **homozygous variant** (two copies of the allele).

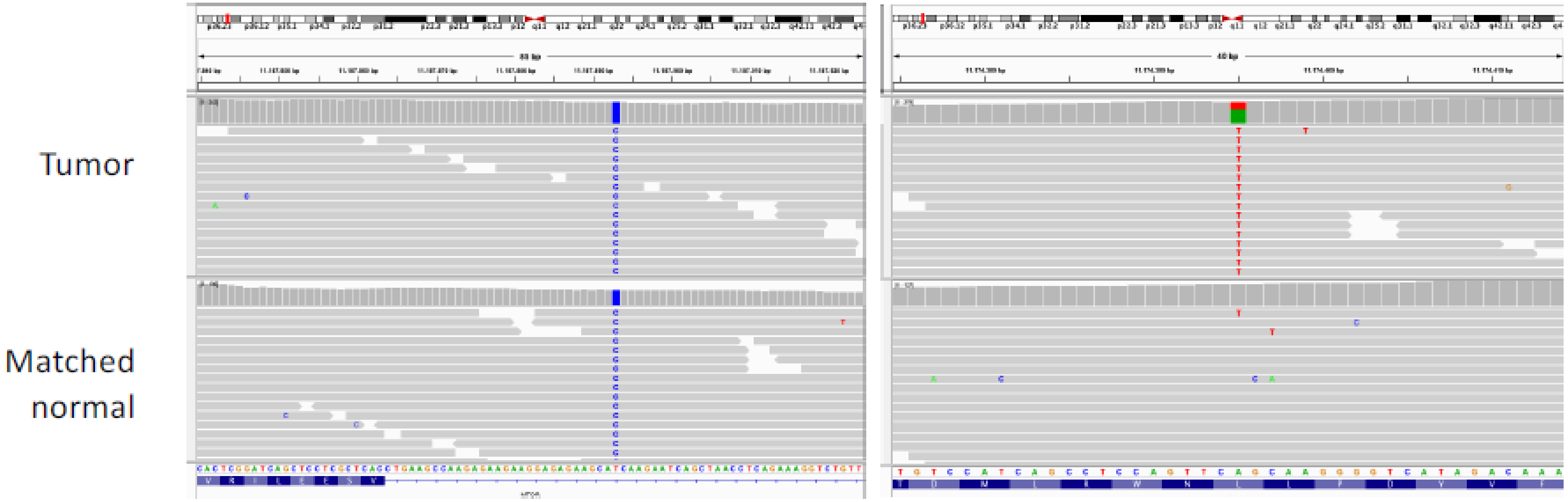
- **Input Data**: Typically uses DNA from a single healthy tissue sample (e.g., blood).
- **Core Assumption**: The variants are constitutional, meaning they are present in all cells.
- **Example Tool**: GATK's HaplotypeCaller is a widely used algorithm specifically designed for this purpose.

Somatic Variant Calling

Somatic variant calling is significantly **more complex** because it's looking for **changes** that are **present in only a subset of the sequenced cells**. For example, in a tumor sample, the cancer cells with the mutation are mixed with normal, non-cancerous cells. This means the variant allele frequency can be very low, well below the 50% or 100% threshold of germline variants.

- **Input Data**: Better using matched "**tumor-normal**" pair. The tumor sample is the primary source of the cancer variants, while a normal sample (e.g., blood or adjacent tissue) from the same individual serves as a crucial control to filter out germline variants.
- **Core Assumption**: Variants are sub-clonal and may be present at very low frequencies, so the signal must be distinguished from random sequencing noise.
- **Key Challenge**: Differentiating a low-frequency somatic variant from a low-quality sequencing artifact is a major statistical hurdle.
- **Example Tool**: Mutect2 is a common algorithm designed to handle this complexity by comparing the tumor and normal samples to identify changes unique to the tumor.

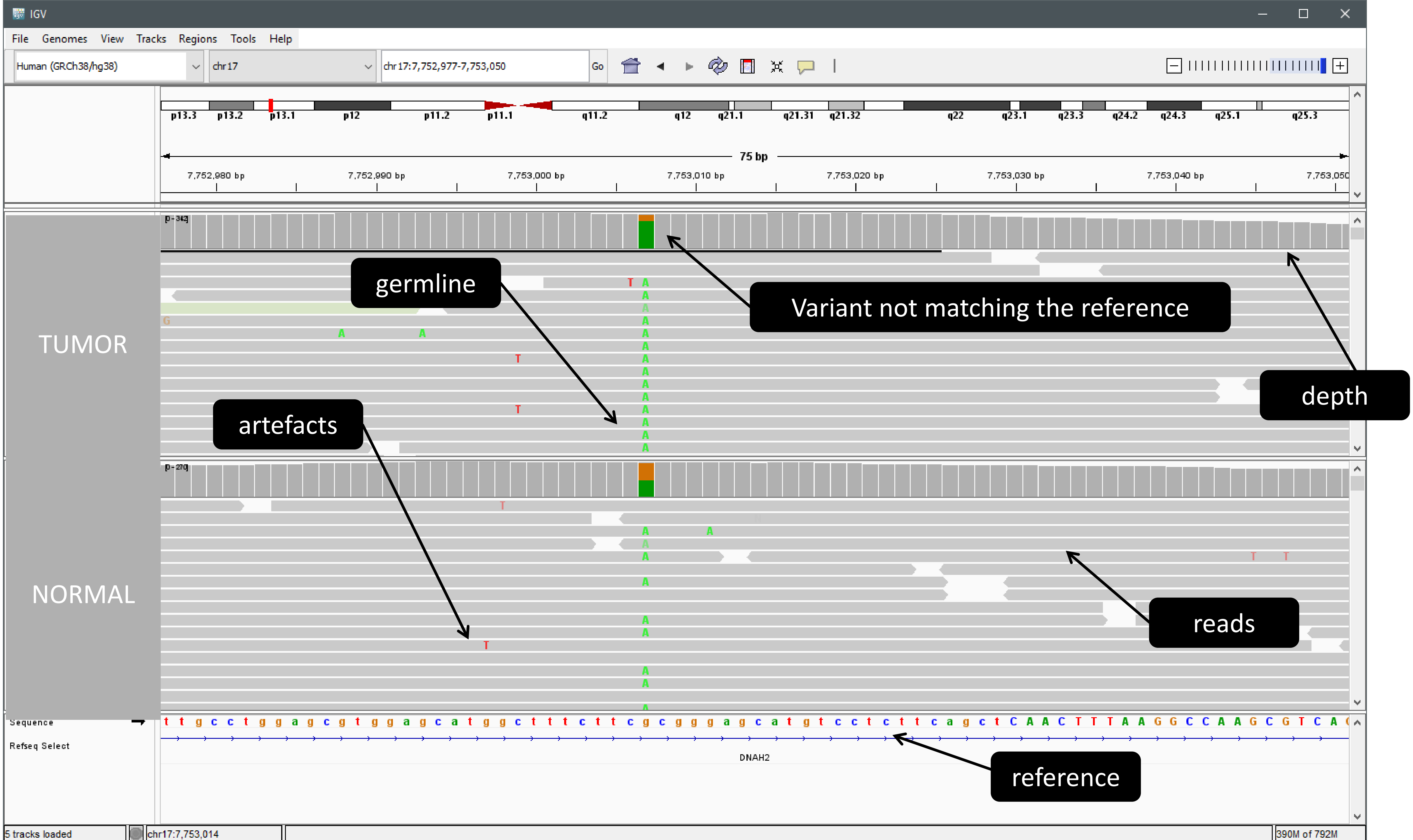
Sequencing both tumor and matched normal samples is best for the detection of somatic mutations

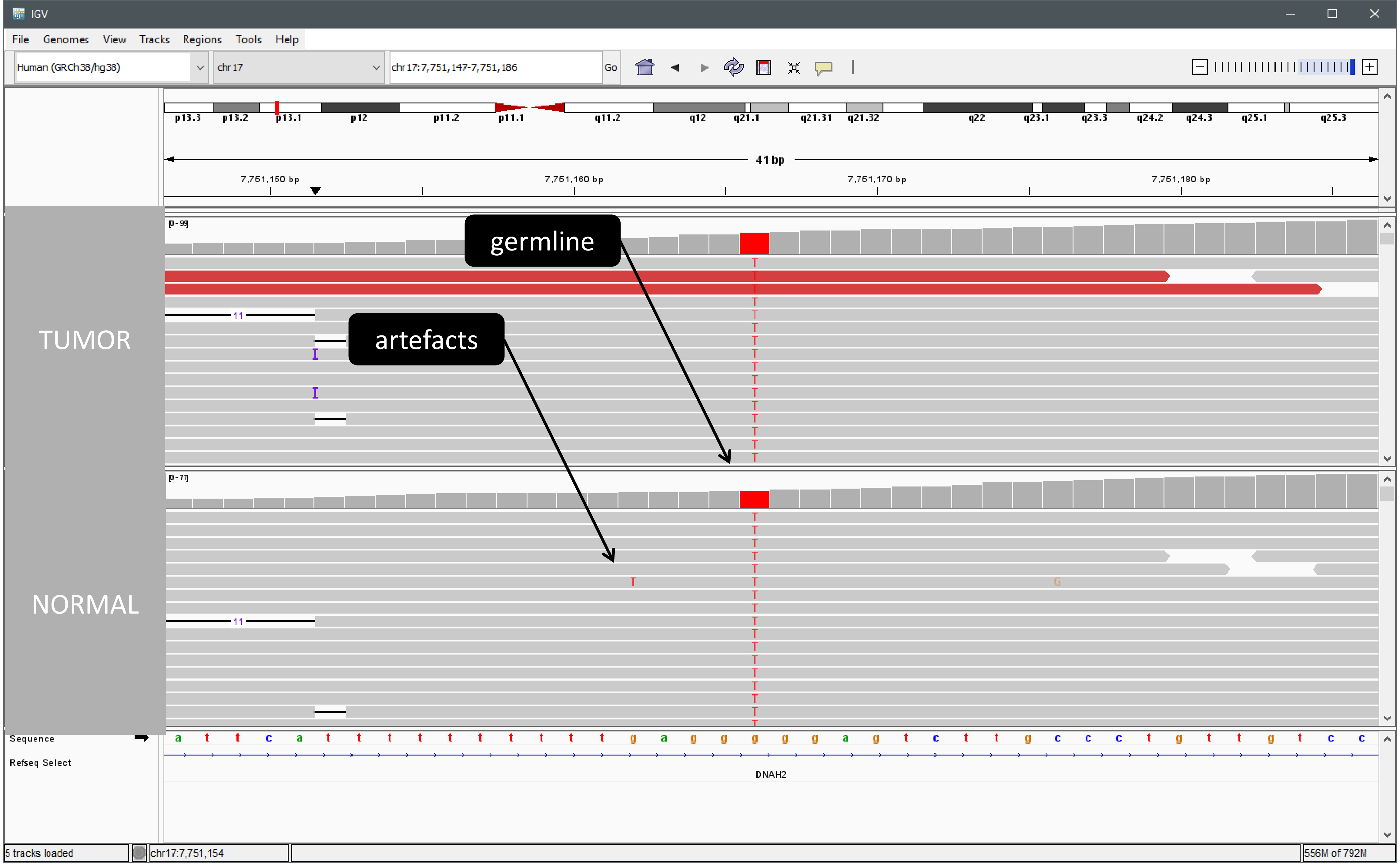


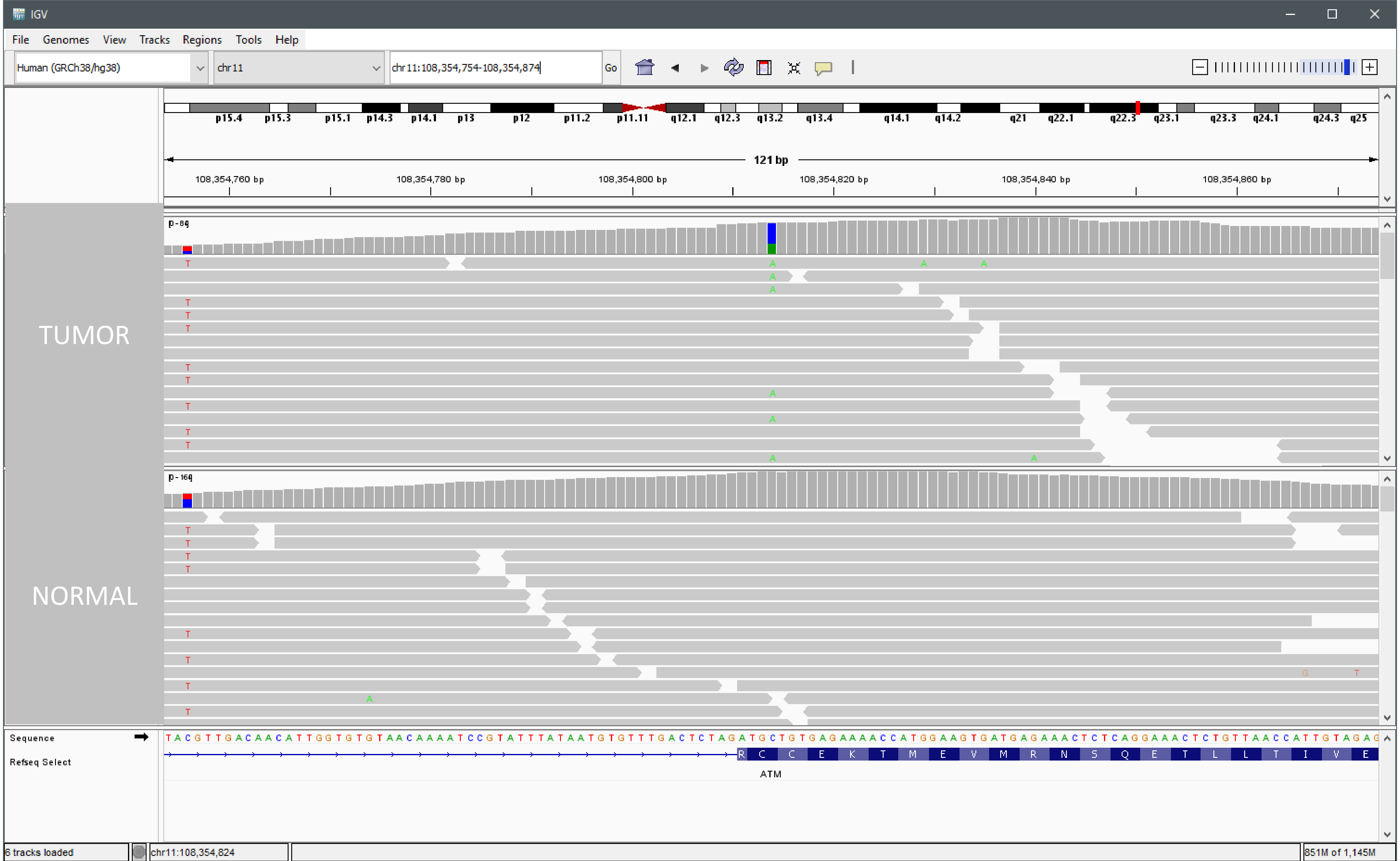
tumor and matched normal sample → somatic variant caller jointly tests T and N

tumor only → somatic variant caller jointly tests T and (pool of) unmatched N

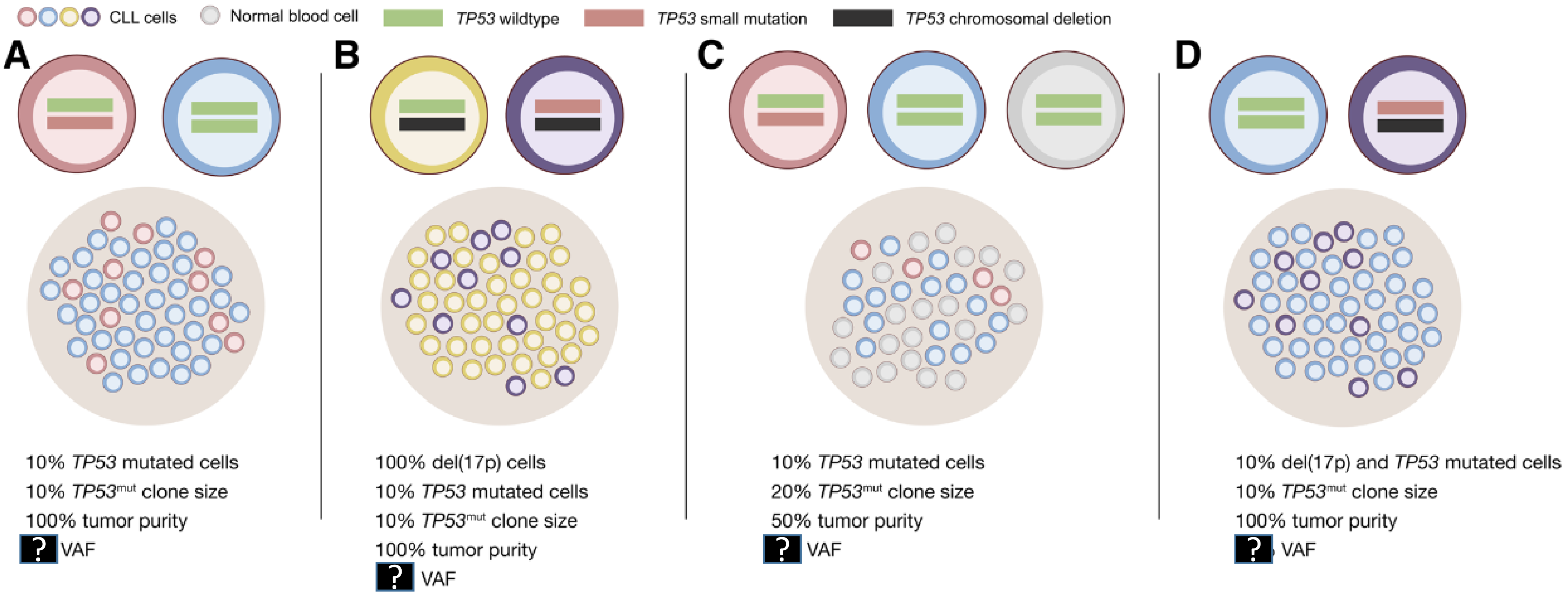
→ single sample somatic variant caller test T alone


IGV
File Genomes View Tracks Regions Tools Help
Human (GRCh38/hg38)
chr17
chr17:7,752,977-7,753,050
Go
75 bp
7,752,980 bp
7,752,990 bp
7,753,000 bp
7,753,010 bp
7,753,020 bp
7,753,030 bp
7,753,040 bp
TUMOR
NORMAL
germline
Variant not matching the reference
depth
artefacts
reads
reference
Sequence
Refseq Select
DNAH2
5 tracks loaded
chr17:7,753,014
390M of 792M


IGV
File
Genomes
View
Tracks
Regions
Tools
Help
Human (GRCh38/hg38)
chr17
chr17:7,751,147-7,751,186
Go
p13.3 p13.2 p13.1 p12 p11.2 p11.1 q11.2 q12 q21.1 q21.31 q21.32 q22 q23.1 q23.3 q24.2 q24.3 q25.1 q25.3
41 bp
7,751,150 bp
7,751,160 bp
7,751,170 bp
7,751,180 bp
[0-99]
germline
artefacts
TUMOR
[0-77]
NORMAL
Sequence
Refseq Select
DNAH2
5 tracks loaded
chr17:7,751,154
556M of 792M


IGV
File Genomes View Tracks Regions Tools Help
Human (GRCh38/hg38)
chr11
chr11:108,354,754-108,354,874
Go
p15.4 p15.3 p15.1 p14.3 p14.1 p13 p12 p11.2 p11.11 q12.1 q12.3 q13.2 q13.4 q14.1 q14.2 q21 q22.1 q22.3 q23.1 q23.3 q24.1 q24.3 q25
121 bp
108,354,760 bp
108,354,780 bp
108,354,800 bp
108,354,820 bp
108,354,840 bp
108,354,860 bp
TUMOR
NORMAL
Sequence
Refseq Select
R C C E K T M E V M R N S Q E T L L T I V E
ATM
6 tracks loaded
chr11:108,354,824
851M of 1,145M

Variable VAF in tumor



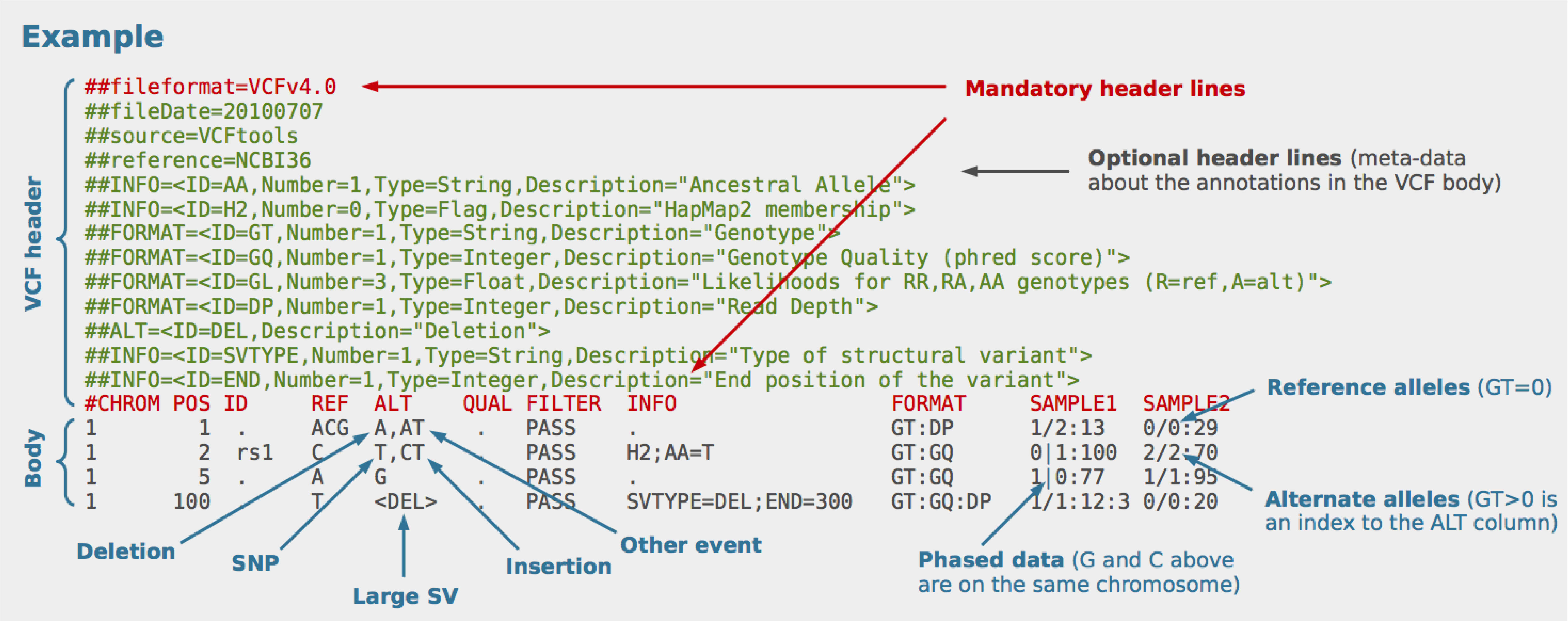
One cannot simply look at a **VAF** and know the **clone size** or vice-versa. You must account for **tumor purity** and the **copy number** of the gene to accurately interpret your results.

Pandzic T et al, Hemasphere. 2022

SECTION BREAK

VCF File Structure

```
##fileformat=VCFv4.1
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS     ID         REF    ALT     QUAL FILTER INFO                              FORMAT      NA00001        NA00002        NA00003
20     14370   rs6054257  G      A       29   PASS   NS=3;DP=14;AF=0.5;DB;H2           GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51 1/1:43:5:.,.
20     17330   .          T      A       3    q10    NS=3;DP=11;AF=0.017               GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3   0/0:41:3
20     1110696 rs6040355  A      G,T     67   PASS   NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2   2/2:35:4
20     1230237 .          T      .       47   PASS   NS=3;DP=13;AA=T                   GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51 0/0:61:2
20     1234567 microsat1  GTC    G,GTCT  50   PASS   NS=3;DP=9;AA=G                    GT:GQ:DP    0/1:35:4       0/2:17:2       1/1:40:3
```



https://davetang.github.io/learning_vcf_file/img/vcf_format.png

HEADER

```
##fileformat=VCFv4.1
##fileDate=20090805
##tcgaversion=1.1
##vcfProcessLog=<InputVCF=<file1.vcf>,InputVCFSource=<caller1>,InputVCFVer=<1.0>,InputVCFParam=<a1,b>,InputVCFgeneAnno=<anno1.gaf>>
##reference=ftp://ftp.ncbi.nih.gov/genbank/genomes/Eukaryotes/vertebrates_mammals/Homo_sapiens/GRCh37/special_requests/GRCh37-lite.fa
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
```

INFO meta-information

```
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
```

FILTER meta-information

```
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
```

FORMAT meta-information

```
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
```

```
##SAMPLE=<ID=NORMAL,Individual=TCGA-01-1000,File=TCGA-01-1000-1.bam,Platform=Illumina,Source=dbGAP,Accession=1234>
##SAMPLE=<ID=TUMOR,Individual=TCGA-01-1000,File=TCGA-01-1000-2.bam,Platform=Illumina,Source=dbGAP,Accession=4567>
##PEDIGREE=<Name_0=TUMOR,Name_1=NORMAL>
```

Fixed fields

Optional: FORMAT field specifying data type + Per-sample genotype data

BODY

```
#CHROM POS      ID         REF   ALT      QUAL FILTER INFO                           FORMAT      NORMAL          TUMOR
20     14370    rs6054257  G     A        29   PASS   NS=3;DP=14;AF=0.5;DB;H2        GT:GQ:DP:HQ 0|0:48:1:51,51  1|0:48:8:51,51
20     17330    .          T     A        3    q10    NS=3;DP=11;AF=0.017            GT:GQ:DP:HQ 0|0:49:3:58,50  0|1:3:5:65,3
20     1110696  rs6040355  A     G,T      67   PASS   NS=2;DP=10;AF=0.333,0.667;DB   GT:GQ:DP:HQ 1|2:21:6:23,27  2|1:2:0:18,2
20     1230237  .          T     .        47   PASS   NS=3;DP=13;AA=T                GT:GQ:DP:HQ 0|0:54:7:56,60  0|0:48:4:51,51
20     1234567  microsat1  GTC   G,GTCTC  50   PASS   NS=3;DP=9;AA=G                 GT:GQ:DP    0/1:35:4        0/2:17:2
```

Genotype

```
FORMAT      NA00001
GT:GQ:DP:HQ 0|0:48:1:51,51
GT:GQ:DP:HQ 0|0:49:3:58,50
GT:GQ:DP:HQ 1|2:21:6:23,27
GT:GQ:DP:HQ 0|0:54:7:56,60
GT:GQ:DP    0/1:35:4
```

vcf

Homozygous Reference (HOM-REF)

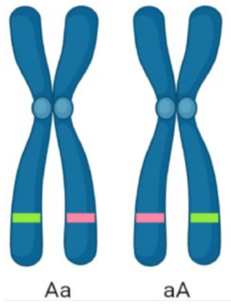
Definition: *The individual has two copies of the reference allele at a particular genetic locus.*

Meaning: *The individual's genotype matches the reference genome at that position, and there is no known variation.*

Heterozygous (HET)

Definition: *The individual has two different alleles at a particular locus.*

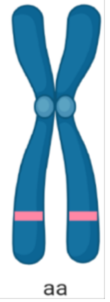
Meaning: *One allele is the reference allele, and the other is an alternative allele.*



Homozygous Alternate (HOM-ALT)

Definition: *The individual has two copies of the alternative allele at a particular genetic locus.*

Meaning: *Both alleles are different from the reference, representing a variant that is present in both copies.*

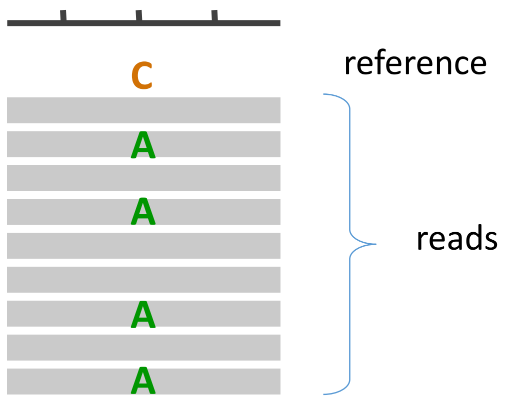


SECTION BREAK

Genotype calculation (germline)

The Model: Binomial Distribution

- **Number of trials** (n)**:** The total number of reads sequenced at this position. Here, n = 9 = 5(ref) + 4(alt)
- **Number of successes** (k)**:** The number of times we observe the alternate allele. Here, k=4.
- **Probability of success** (p)**:** The theoretical probability of observing the alternate allele, given the model. For a diploid, heterozygous individual, we have one reference allele and one alternate allele, so the theoretical probability is *p=0.5*.

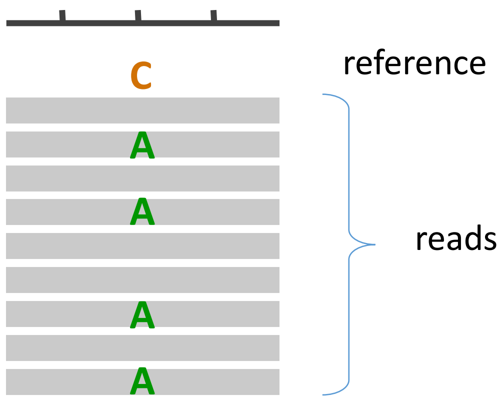


The formula for the binomial probability is:

$$P(X = k) = \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k} = \frac{n!}{k!(n-k)!} \cdot p^k \cdot (1-p)^{n-k}$$

Binomial Distribution

$$P(X=k)=\binom{n}{k}\cdot p^k\cdot(1-p)^{n-k}=\frac{n!}{k!(n-k)!}\cdot p^k\cdot(1-p)^{n-k}$$



What is the chance of seeing exactly 4 alternate reads out of 9 total reads, if the true genotype is heterozygous?

$$P(X=4)=\binom{9}{4}\cdot 0.5^4\cdot(1-0.5)^{9-4}=126\cdot 0.5^4\cdot 0.5^5\approx 0.246$$

This is the probability of the observed data given the assumed model.

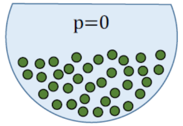
The probability of observing 4 alternate reads out of 9, assuming a true heterozygous variant, is approximately 24.6%.

The probability of observing 4 alternate reads out of 9, assuming a hom. ref?

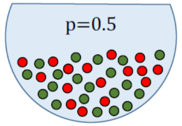
The probability of observing 4 alternate reads out of 9, assuming a hom. alt?

Likelihood

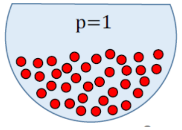
In this context, **likelihood is the probability of the model given the observed data**. The numerical value is the same as the probability we just calculated, but the interpretation is different. While probability answers, "*What is the chance of this data occurring*?" (given a known model), likelihood answers, "*How well does this model explain the data*?" (when the model is unknown).
Variant callers don't just calculate one likelihood; they calculate the likelihood for multiple possible genotypes (models) and then pick the one with the highest value.



- **Likelihood (Homozygous Reference):** What is the chance of observing 4 alternate reads if the true genotype is reference/reference? (p = 0.0) -> Likelihood is 0.



- **Likelihood (Heterozygous):** What is the chance of observing 4 alternate reads if the true genotype is heterozygous? (p = 0.5) -> Likelihood is approximately 0.246.



- **Likelihood (Homozygous Alternate):** What is the chance of observing 4 alternate reads if the true genotype is alternate/alternate? (p = 1.0) -> Likelihood is 0.

In this case, the likelihood of the heterozygous model being the correct one, given the data, is 0.246. Since this is the highest likelihood among the common genotypes, the variant caller would likely conclude that the genotype is indeed heterozygous.

The formula for likelihood is the same mathematical function as the probability formula, but the interpretation and what you are solving for are different. The general formula for the likelihood of a parameter θ, given some observed data x, is:

$$L(\theta|x) = f(x|\theta)$$

probability mass function (or density function) that you use to calculate the probability of the data. For variant calling, this is the binomial probability formula.

likelihood function. It represents how plausible a specific model parameter ϑ is, given your observed data *x*.

Applying the Formula to Variant Calling

In your example, the model parameter ϑ is the true allele frequency (*p*) of the variant. The observed data *x* is the number of alternate reads (*k*) out of the total number of reads (*n*). So, the likelihood formula for a variant call is the same as the binomial probability formula:

$$L(p|k,n) = \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k}$$

Estimating the Genotype

This formula calculates the likelihood of a given genotype based on the *observed reads.*

It adds a crucial **layer of complexity** by explicitly modeling the sequencing error rate (ϵ).

$$\mathcal{L}(g) = \frac{1}{m^k} \prod_{j=1}^{l} [(m-g)\epsilon_j + g(1-\epsilon_j)] \prod_{j=l+1}^{k} [(m-g)(1-\epsilon_j) + g\epsilon_j]$$

where

$\mathcal{L}(g)$ *likelihood* of the assumed genotype g. This is the value we're trying to find
m *ploidy.* For a human genome, this is almost always m=2 (diploid)
k *total number of reads* at this position
l *number of reads with the reference allele*
$k-l$ *number of reads with the alternate allele*
ϵ *sequencing error rate*. This is the probability that a single sequenced base is incorrect.
g *count of ref alleles (g=2 -> Ref/Ref)*

The Logic of the Formula

The formula is a product of probabilities, meaning it calculates the likelihood by multiplying the individual probabilities of each observed read. It's essentially saying: "What is the probability of observing this exact set of k reads, given my assumption about the true genotype (g)?".The formula can be broken down into three main parts:

$$\mathcal{L}(g) = \frac{1}{m^k} \underbrace{\prod_{j=1}^{l} [(m-g)\epsilon_j + g(1-\epsilon_j)]}_{\text{product for reference reads}} \underbrace{\prod_{j=l+1}^{k} [(m-g)(1-\epsilon_j) + g\epsilon_j]}_{\text{product for alternative reads}}$$

A

B

This term is a normalization constant that is often dropped in practice because it is constant for a given position and doesn't change the relative likelihoods of different genotypes. It ensures the sum of all probabilities equals one, but it's not the core of the biological calculation.

A For each reference read: The formula asks, "What's the probability of this being a reference read?" It could be a correct read from the reference allele (99% chance), or a sequencing error on the alternate allele (1% chance). The formula combines these.

B For each alternate read: The formula asks, "What's the probability of this being an alternate read?" It could be a correct read from the alternate allele (99% chance), or a sequencing error on the reference allele (1% chance).

$$\mathrm{PL} = -10\log_{10}(\mathcal{L}(g))$$

Genotype Likelihood Calculations

Total Reads (k): 9, Ref Reads (l): 8, Alt Reads: 1

Error Rate (e): 0.01

Likelihood (g=0, Homozygous Ref): 9.2274e-03

PL Score (g=0): 20.35

Likelihood (g=1, Heterozygous): 1.9531e-03

PL Score (g=1): 27.09

Likelihood (g=2, Homozygous Alt): 9.9000e-17

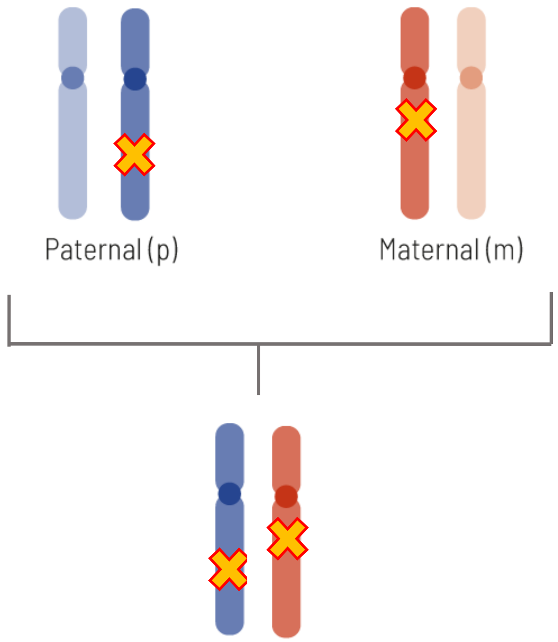
PL Score (g=2): 160.04

Note: The normalization term 1/m^k is omitted as it does not affect relative likelihoods.

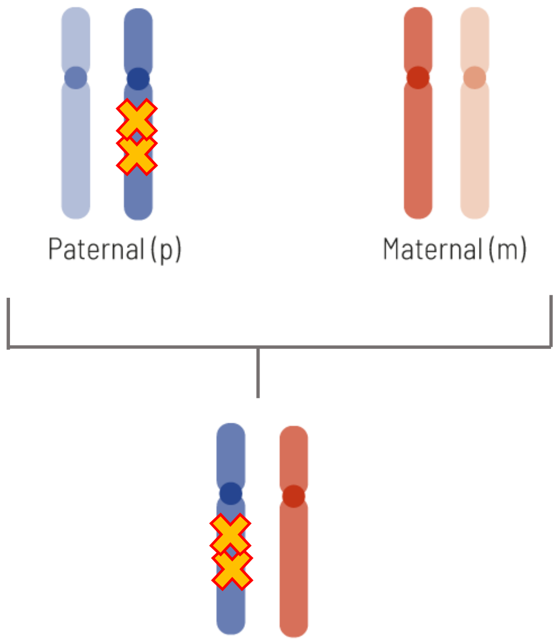
Phased variants

```
##fileformat=VCFv4.1
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS     ID         REF    ALT     QUAL FILTER INFO                              FORMAT      NA00001        NA00002        NA00003
20     14370   rs6054257  G      A       29   PASS   NS=3;DP=14;AF=0.5;DB;H2           GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51 1/1:43:5:.,.
20     17330   .          T      A       3    q10    NS=3;DP=11;AF=0.017               GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3   0/0:41:3
20     1110696 rs6040355  A      G,T     67   PASS   NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2   2/2:35:4
20     1230237 .          T      .       47   PASS   NS=3;DP=13;AA=T                   GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51 0/0:61:2
20     1234567 microsat1  GTC    G,GTCT  50   PASS   NS=3;DP=9;AA=G                    GT:GQ:DP    0/1:35:4       0/2:17:2       1/1:40:3
```

Phasing

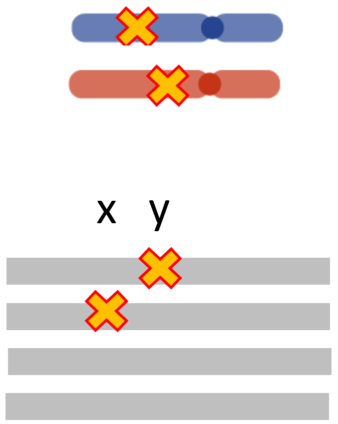


both copies affected
disease

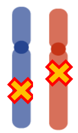


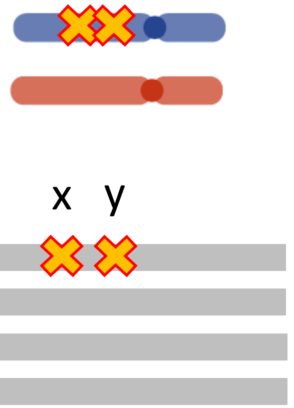
only one copy affected
No disease

Phasing

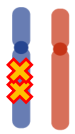


CHROM	POS	ID	S1
chr20	101	snp_x	0|1
chr20	204	snp_y	1|0





CHROM	POS	ID	S1
chr20	101	snp_x	1|0
chr20	204	snp_y	1|0



Variant calling : basic steps

1. Read mapping / preprocessing

- **Goal**: Align raw sequencing reads to a reference genome.

- **Steps**
  - Read mapping (e.g., BWA, Bowtie2).
  - Remove duplicates, perform base quality score recalibration.

- **Stats**
  - Mapping quality (MAPQ) = $-10 \cdot \log_{10} P(\text{read misaligned})$.
  - Base quality (Phred scale) = $-10 \cdot \log_{10} P(\text{base call wrong})$.

2. Candidate variant detection

- **Goal**: Identify positions in the genome where the observed bases differ from the reference at a frequency that cannot be explained by random sequencing errors alone

Statistical approaches used

(a) Base counting and pileup
(b) Binomial test for allele imbalance
(c) Likelihood-based evaluation in Variant Calling

a) Base counting and pileup

- At each genomic position, the caller collects all aligned reads (this is the *pileup*).
- It counts how many reads support the **reference allele** and how many support an **alternative allele.**
- Based on those counts, it calculates an **allele frequency (AF)**, and then applies a statistical test to decide whether the observed difference is likely due to sequencing errors or represents a real variant.

At a given genomic position:

- Let
  - n = total number of reads covering the position
  - r = number of reads supporting the **reference allele**
  - a = number of reads supporting the **alternative allele**
  - q_i = base quality score of read i
- **Allele frequency (AF):**

$$AF = \frac{a}{n}$$

- **Error model (simplified):** Each base has an error probability

$$p_i = 10^{-\frac{Q_i}{10}}$$

where Q_i is the Phred quality score.

- **Likelihood of data given a true genotype G:**

$$L(G) = \prod_{i=1}^{n} P(\text{base}_i \mid G, p_i)$$

For example:

- If genotype = homozygous reference (RR), probability of seeing an alternative read is ~p_i.
- If genotype = heterozygous (RA), probability of seeing ref vs alt is ~0.5 each.
- If genotype = homozygous alt (AA), probability of seeing reference is ~p_i.
- **Genotype calling:** The variant caller compares likelihoods for the possible genotypes (RR, RA, AA) and reports the one with the highest posterior probability (often using Bayes' rule).

🧮 Example

We have **10 reads** covering one position.

- 7 reads show an **alternative allele (A)**
- 3 reads show the **reference allele (G)**
- All bases have **quality score Q30** (so error rate ~0.001 = 0.1%)

---

1. Allele frequency (AF)

$$AF = \frac{a}{n} = \frac{7}{10} = 0.7$$

So 70% of reads support the alternative allele.

3. Compute likelihood for each genotype

We test 3 possible genotypes:

- **RR (homozygous reference)**
- **RA (heterozygous)**
- **AA (homozygous alternative)**

(a) Homozygous reference (RR):

- Expected: almost all reads = reference
- Probability of seeing an *alt read* = error rate = 0.001
- Probability of seeing a *ref read* = 1 – 0.001 = 0.999

$$L(RR) = (0.999)^3 \times (0.001)^7$$

That's an extremely small number (≈ 10^{-21}).

(b) Homozygous alternative (AA):

- Expected: almost all reads = alternative
- Probability of seeing an *alt read* = 0.999
- Probability of seeing a *ref read* = 0.001

$$L(AA) = (0.001)^3 \times (0.999)^7$$

$\approx 10^{-9}$ (still small, but much larger than RR).

(c) Heterozygous (RA):

- Expected: ~50% ref, ~50% alt
- Probability of seeing either ref or alt ≈ 0.5

$$L(RA) = (0.5)^{10} = 0.000976$$

Much larger than RR or AA likelihoods.

4. Compare likelihoods

- $L(RR) \approx 10^{-21}$ → almost impossible
- $L(AA) \approx 10^{-9}$ → possible, but not very likely
- $L(RA) \approx 10^{-3}$ → by far the highest likelihood

👉 **Conclusion:** The caller reports the genotype as **heterozygous (RA)**, with AF ~0.7.

b) Binomial test for allele imbalance

- Suppose at a site you have *n* reads, of which *k* support an alternate allele.
- If sequencing error rate is ϵ, the null hypothesis is:

$$H_0 : k \sim \mathrm{Binomial}(n, \epsilon)$$

- Compute probability of seeing ≥k alternate reads just by error:

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} \epsilon^i (1-\epsilon)^{n-i}$$

- If this probability is very small → site becomes a candidate variant.

The Null Hypothesis ($\mathbf{H_0}$)

The binomial test starts with a very skeptical assumption: **The observed Alt allele count (k) is just random sequencing error.**

- **$\mathbf{H_0}$:** The Alt allele is **not truly present** (the genotype is Homozygous Reference, Ref/Ref). Therefore, every Alt read is simply a **sequencing error** that happened to occur at that position.
- The probability of seeing an Alt allele is the **Sequencing Error Rate (ϵ).**

The Goal

The test's goal is to determine: **How likely is it to observe k or more Alt reads, assuming the only reason they appeared is random sequencing error (ϵ)?**

The Formula

probability of getting **exactly i successful events** (Alt reads):

$$\mathbf{P(X = i)} = \binom{\mathbf{n}}{\mathbf{i}} \cdot \epsilon^{\mathbf{i}} \cdot (\mathbf{1} - \epsilon)^{\mathbf{n-i}}$$

The probability of the remaining $(n - i)$ reads being **Ref reads** (i.e., not errors).

The probability of i independent Alt reads occurring, assuming each one is an **error** (with probability ϵ).

2. The Summation (The p-value)

You are not just interested in the probability of exactly k Alt reads; you want the probability of k **or more** Alt reads.

$$\mathbf{P(X \geq k) = \sum_{i=k}^{n} \dots}$$

- This is known as the **p-value** (specifically, the **one-sided upper tail** p-value).
- The summation adds up the probabilities for every possible outcome that is **as extreme as, or more extreme than,** what you actually observed (from your observed count k up to the maximum possible count n).

Example: Putting it Together

Imagine you have a depth of $n = 10$ reads, you observe $k = 3$ Alt reads, and the known error rate is $\epsilon = 0.01$ (1%).

Question: How likely is it to see 3 or more Alt reads if the true genotype is Ref/Ref (i.e., only error)?

$$\mathbf{P(X \geq 3)} = P(X = 3) + P(X = 4) + \cdots + P(X = 10)$$

- $P(X = 3) = \binom{10}{3} \cdot (0.01)^3 \cdot (0.99)^7 \approx 0.00011$
- $P(X = 4)$ would be even smaller, and so on.

The sum of these probabilities is your p-value.

- If the $\mathbf{p-value}$ is very **small** (e.g., $p < 0.001$), you **reject $\mathbf{H_0}$**. This means the chance of seeing 3 Alt reads due to error alone is tiny. You conclude the Alt allele is **truly present** (a real variant).
- If the $\mathbf{p-value}$ is **large** (e.g., $p > 0.05$), you **fail to reject $\mathbf{H_0}$**. You conclude the 3 Alt reads are **consistent with random error** and the true genotype is likely Ref/Ref.

5. Filtering & post-processing

- **Goal**: Reduce false positives.
- **Stats:**
  - **Hard filters**: thresholds on depth, strand bias, mapping quality, allele fraction.
  - **Strand bias tests**: Fisher’s exact test or binomial test.
  - **Overdispersion modeling**: beta-binomial (to model extra variance in allele counts).
  - **Machine learning** (GATK VQSR): Gaussian mixture models to separate true vs false variants using training sets.

end